

ECON 672 — Metrics

Expanded Notes

1 Generalized Method of Moments

1.1 Introduction

The Generalized Method of Moments (GMM) is one of the central estimation frameworks in econometrics. Its importance comes from two related facts. First, it generalizes the classical Method of Moments (MM). Second, it provides a common language for studying many estimators that may at first look unrelated, including ordinary least squares (OLS) and instrumental variables (IV).

At a broad level, the logic of GMM is simple. Economic models often imply that, at the true parameter value, certain population moments must equal zero. The econometrician then chooses the parameter value that makes the corresponding sample moments as close to zero as possible. In this sense, GMM turns theoretical restrictions from the model into an estimation procedure.

Why GMM matters

GMM is not just one more estimator. It is a unifying framework. Many familiar procedures in econometrics can be written as estimators that exploit moment restrictions. This is why GMM appears, either explicitly or implicitly, in a very large share of empirical work in economics.

1.2 References for this Unit

The main references for this topic are Hayashi (2000) and Hansen (2022). For empirical applications and intuition, useful references include Angrist and Krueger (1991) and Angrist and Pischke (2015).

Remark. A good way to think about these references is the following. Hayashi provides the general econometric framework and the formal theory of GMM. Hansen emphasizes the probabilistic and statistical foundations. Angrist and Krueger, together with Angrist and Pischke, show how these ideas appear in empirical practice, especially in instrumental variables applications.

1.3 From Method of Moments to GMM

Suppose the data are summarized by a random vector W_i , and let $\theta_0 \in \Theta \subseteq \mathbb{R}^k$ denote the true parameter vector. A model delivers a vector of population moment conditions of the form

$$\mathbb{E}[g(W_i, \theta_0)] = 0,$$

where $g(W_i, \theta)$ is an $m \times 1$ vector of moment functions.

The basic Method of Moments idea is to replace the population condition by its sample analogue:

$$\frac{1}{n} \sum_{i=1}^n g(W_i, \theta) \approx 0,$$

and then choose θ so that these sample moments are equal, or as close as possible, to zero.

This immediately raises an important question. What should be done when there are more moment conditions than parameters, that is, when $m > k$? In that case, one typically cannot solve

$$\frac{1}{n} \sum_{i=1}^n g(W_i, \theta) = 0$$

exactly. GMM resolves this problem by choosing θ to minimize a weighted quadratic form in the sample moments. This is the key generalization from MM to GMM.

Definition. A *moment condition* is a population restriction of the form

$$\mathbb{E}[g(W_i, \theta_0)] = 0.$$

An estimator based on such restrictions uses the sample analogue of these moments to learn about the unknown parameter θ_0 .

1.4 GMM as a Unifying Framework

One reason GMM is so useful is that many standard estimators can be written in moment form.

Example (OLS as a moment-based estimator). Consider the linear regression model

$$Y_i = X_i' \beta_0 + u_i,$$

with the exogeneity condition

$$\mathbb{E}[X_i u_i] = 0.$$

Since $u_i = Y_i - X_i' \beta_0$, this restriction can be written as

$$\mathbb{E}[X_i (Y_i - X_i' \beta_0)] = 0.$$

Thus OLS can be viewed as an estimator based on a moment condition.

Example (IV as a moment-based estimator). In an instrumental variables setting, suppose Z_i is a vector of instruments satisfying

$$\mathbb{E}[Z_i u_i] = 0.$$

Then the identifying restriction becomes

$$\mathbb{E}[Z_i(Y_i - X_i' \beta_0)] = 0.$$

This is again a moment condition, now based on instruments rather than regressors. Hence IV also fits naturally into the GMM framework.

These examples illustrate the main conceptual point: GMM does not replace OLS or IV as separate objects, but instead shows that both can be understood as special cases of a more general estimation strategy.

1.5 Why Economists Use GMM So Often

GMM is used widely because it is both flexible and theory-driven. It is flexible because it allows the econometrician to work directly with the implications of a model, without necessarily specifying the full distribution of the data. It is theory-driven because the estimator is built from restrictions that come from economic structure, statistical assumptions, or both.

This combination is especially powerful in economics. In many applications, the model implies valid orthogonality conditions even when a full likelihood is difficult to specify or unattractive to impose. GMM provides a systematic way to exploit exactly that information.

Remark. In practice, much of econometrics can be viewed as the study of how to construct, justify, and use moment restrictions. From this perspective, GMM is not a niche technique but one of the main organizing principles of modern econometric theory.

1.6 Setup

We now consider a general class of *moment equation models*. The parameter of interest is

$$\beta \in B \subseteq \mathbb{R}^k,$$

where k is the dimension of the unknown parameter vector. The model is characterized by a collection of population moment restrictions of the form

$$\mathbb{E}[g_i(\beta_0)] = 0,$$

where $g_i(\beta)$ is a known $\ell \times 1$ function of β , and β_0 denotes the true parameter value.

The key idea is that the economic or statistical model implies that, at the true parameter, certain moments must vanish. Estimation will therefore proceed by searching for values of β

that make the sample analogue of these moments close to zero.

Definition. A *moment equation model* is a model in which the true parameter β_0 satisfies

$$\mathbb{E}[g_i(\beta_0)] = 0,$$

for some known vector-valued function $g_i(\beta) \in \mathbb{R}^\ell$.

This framework is very general and nests many models we have already studied.

Example (Linear regression as a moment condition model). Consider the linear model

$$Y_i = X_i' \beta_0 + \varepsilon_i,$$

together with the exogeneity condition

$$\mathbb{E}[X_i \varepsilon_i] = 0.$$

Since $\varepsilon_i = Y_i - X_i' \beta_0$, this can be rewritten as

$$\mathbb{E}[X_i (Y_i - X_i' \beta_0)] = 0.$$

Hence this model fits the moment condition framework with

$$g_i(\beta) = X_i (Y_i - X_i' \beta).$$

Example (Linear IV model as a moment condition model). Consider again

$$Y_i = X_i' \beta_0 + \varepsilon_i,$$

but now suppose X_i may be endogenous and that we have an instrument vector Z_i satisfying

$$\mathbb{E}[Z_i \varepsilon_i] = 0.$$

Then the relevant moment condition becomes

$$\mathbb{E}[Z_i (Y_i - X_i' \beta_0)] = 0,$$

so the model fits the same framework with

$$g_i(\beta) = Z_i (Y_i - X_i' \beta).$$

These examples already show why GMM is so useful: it provides a single language in which both OLS-type and IV-type estimators can be analyzed.

1.7 Identification and the Number of Moments

Because $g_i(\beta)$ is an $\ell \times 1$ vector, the model provides ℓ population moment equations. The parameter vector β has dimension k . The relation between ℓ and k is therefore fundamental.

Definition. The model is said to be

- *overidentified* if $\ell > k$,
- *just-identified* if $\ell = k$,
- *underidentified* if $\ell < k$.

A necessary condition for identification is

$$\ell \geq k.$$

That is, the number of moment conditions must be at least as large as the number of unknown parameters.

Remark. This condition is only necessary, not sufficient. Having at least as many moment equations as unknown parameters does not by itself guarantee that β_0 is uniquely pinned down by the observable distribution of the data. The moment conditions must also be informative enough to distinguish different parameter values.

Remark. Intuitively, each moment condition contributes one restriction that the true parameter must satisfy. If there are fewer restrictions than unknown parameters, then one should generally not expect the data to determine β_0 uniquely.

1.8 Linear and Nonlinear Models

The GMM framework is broad enough to accommodate both linear and nonlinear models. This is one of its main strengths.

If the moment function $g_i(\beta)$ is linear in β , the resulting estimators often admit explicit closed-form solutions. This makes both computation and theoretical analysis substantially simpler. For that reason, we will mostly focus on linear GMM estimators.

In the linear case that will be most relevant for us,

$$g_i(\beta) = Z_i(Y_i - X_i'\beta).$$

This includes the standard linear IV setup as a special case.

Remark. When the moment conditions are linear in the parameter, the GMM estimator can often be written in an explicit algebraic form, much like OLS or IV. This is why linear GMM is a natural starting point for theory.

Nonlinear GMM is conceptually similar, but its analysis requires additional tools. In particular, nonlinear moment conditions are often studied by using local approximations, such as Taylor expansions, to relate them to linear problems. This step is powerful, but it introduces additional technical subtleties. For that reason, nonlinear GMM is best treated separately and more carefully.

Big picture

The general GMM framework starts from moment restrictions

$$\mathbb{E}[g_i(\beta_0)] = 0.$$

The object k is the number of unknown parameters, while ℓ is the number of moment equations. Linear regression and IV both fit naturally into this framework, and linear GMM will be our main focus because it already captures the essential ideas while remaining algebraically tractable.

1.9 An Euler Equation Example

To illustrate how general the GMM framework is, consider a standard asset-pricing Euler equation, following Hayashi (2000) and Hansen and Singleton (1982). The key population restriction is

$$\mathbb{E} \left(R_{t+1} \beta_0 \frac{u'(C_{t+1})}{u'(C_t)} \middle| \mathcal{I}_t \right) = 1,$$

where:

- R_{t+1} is the gross ex-post rate of return between periods t and $t + 1$,
- β_0 is the subjective discount factor,
- C_t and C_{t+1} denote consumption at dates t and $t + 1$,
- $\mathcal{I}_t$ is the information set available at time t .

Equivalently,

$$\mathbb{E} \left(R_{t+1} \beta_0 \frac{u'(C_{t+1})}{u'(C_t)} - 1 \middle| \mathcal{I}_t \right) = 0.$$

Remark. This is a conditional moment restriction. It says that, given the information available at time t , the marginal utility-adjusted return has conditional expectation equal to one.

1.9.1 CRRA Utility

Suppose utility takes the constant relative risk aversion (CRRA) form

$$u(c) = \frac{c^{1-\alpha_0}}{1-\alpha_0}.$$

Then

$$u'(c) = c^{-\alpha_0}.$$

Substituting this into the Euler equation yields

$$\mathbb{E} \left(R_{t+1} \beta_0 \left(\frac{C_{t+1}}{C_t} \right)^{-\alpha_0} - 1 \middle| \mathcal{I}_t \right) = 0. \quad (1)$$

Thus the unknown parameter vector is

$$\beta = (\alpha_0, \beta_0)'.$$

1.9.2 From Conditional to Unconditional Moments

Let $X_t \in \mathcal{I}_t$, where X_t is any variable measurable with respect to the information set at time t . Then equation (1) implies

$$\mathbb{E} \left[X_t \left(R_{t+1} \beta_0 \left(\frac{C_{t+1}}{C_t} \right)^{-\alpha_0} - 1 \right) \right] = 0$$

by the Law of Iterated Expectations.

Remark. This is the key step that connects economic theory to GMM. The model first gives a *conditional* moment restriction. By multiplying by any variable known at time t and applying the Law of Iterated Expectations, we obtain *unconditional* moment restrictions, which are exactly the objects used in GMM estimation.

To see this formally, define

$$m_{t+1}(\beta) = R_{t+1} \beta_0 \left(\frac{C_{t+1}}{C_t} \right)^{-\alpha_0} - 1.$$

Then the conditional Euler equation says

$$\mathbb{E}[m_{t+1}(\beta_0) \mid \mathcal{I}_t] = 0.$$

If $X_t \in \mathcal{I}_t$, then

$$\mathbb{E}[X_t m_{t+1}(\beta_0)] = \mathbb{E}[\mathbb{E}[X_t m_{t+1}(\beta_0) \mid \mathcal{I}_t]].$$

Since X_t is $\mathcal{I}_t$ -measurable,

$$\mathbb{E}[X_t m_{t+1}(\beta_0) \mid \mathcal{I}_t] = X_t \mathbb{E}[m_{t+1}(\beta_0) \mid \mathcal{I}_t] = 0,$$

and therefore

$$\mathbb{E}[X_t m_{t+1}(\beta_0)] = 0.$$

1.9.3 Moment Function

This fits directly into the GMM framework by defining

$$g_t(\beta) = X_t \left(R_{t+1} \beta_0 \left(\frac{C_{t+1}}{C_t} \right)^{-\alpha_0} - 1 \right),$$

where

$$\beta = (\alpha_0, \beta_0)'$$

Hence the model implies the unconditional moment restriction

$$\mathbb{E}[g_t(\beta_0)] = 0.$$

Interpretation

This example is important because it shows that GMM is not restricted to linear regression settings. Starting from an economic model, we derive equilibrium restrictions in expectation form, convert them into unconditional moments using variables known at time t , and then estimate the structural parameters by matching those moments in the data.

Remark. If X_t is a vector of dimension ℓ , then this construction generates ℓ moment conditions. Since here the unknown parameter vector has dimension $k = 2$, the model is overidentified whenever $\ell > 2$, just-identified when $\ell = 2$, and underidentified when $\ell < 2$.

1.10 The Just-Identified Case: Method of Moments

Recall that the model is characterized by population moment restrictions

$$\mathbb{E}[g_i(\beta_0)] = 0,$$

where $g_i(\beta) \in \mathbb{R}^\ell$ and $\beta_0 \in \mathbb{R}^k$. The sample analogue of the population moment is

$$\bar{g}_n(\beta) = \frac{1}{n} \sum_{i=1}^n g_i(\beta).$$

Consider first the special case in which the model is *just-identified*, so that

$$\ell = k.$$

If the model is identified and the sample moments are well behaved, the natural estimator is the *Method of Moments* (MM) estimator, defined as the solution to

$$\bar{g}_n(\hat{\beta}_{mm}) = 0.$$

The idea is direct: when the number of sample moment equations equals the number of unknown parameters, one can hope to solve the system exactly.

Definition. In the just-identified case, the *Method of Moments estimator* is any solution $\hat{\beta}_{mm}$ to

$$\frac{1}{n} \sum_{i=1}^n g_i(\beta) = 0.$$

Remark. In a just-identified model, estimation amounts to solving a square system of equations. This is the direct analogue of solving k equations in k unknowns.

1.10.1 OLS as a Method of Moments Estimator

Consider the linear regression model

$$Y_i = X_i' \beta_0 + e_i, \quad \mathbb{E}[X_i e_i] = 0.$$

This implies the moment condition

$$\mathbb{E}[X_i(Y_i - X_i' \beta_0)] = 0.$$

Hence

$$g_i(\beta) = X_i(Y_i - X_i' \beta),$$

and the sample moment is

$$\bar{g}_n(\beta) = \frac{1}{n} \sum_{i=1}^n X_i(Y_i - X_i' \beta).$$

Setting $\bar{g}_n(\beta) = 0$ gives

$$\frac{1}{n} \sum_{i=1}^n X_i(Y_i - X_i' \beta) = 0.$$

In matrix notation, with $Y = (Y_1, \dots, Y_n)'$ and $X = (X_1, \dots, X_n)'$, this becomes

$$\frac{1}{n} X'(Y - X\beta) = 0,$$

or equivalently

$$X'X\beta = X'Y.$$

If $X'X$ is invertible, the solution is

$$\hat{\beta}_{ols} = (X'X)^{-1} X'Y.$$

Thus OLS can be viewed as a Method of Moments estimator.¹

¹There is something beautiful to add here. It is in the appendix 7.1.

1.10.2 Just-Identified IV as a Method of Moments Estimator

Now suppose we have instruments $Z_i \in \mathbb{R}^k$ and the moment condition

$$\mathbb{E}[Z_i(Y_i - X_i'\beta_0)] = 0.$$

Then

$$g_i(\beta) = Z_i(Y_i - X_i'\beta),$$

and the MM estimator solves

$$\frac{1}{n} \sum_{i=1}^n Z_i(Y_i - X_i'\beta) = 0.$$

In matrix form,

$$\frac{1}{n} Z'(Y - X\beta) = 0,$$

so that

$$Z'X\beta = Z'Y.$$

If $Z'X$ is invertible, the just-identified IV estimator is

$$\hat{\beta}_{iv} = (Z'X)^{-1}Z'Y.$$

This is another example of a Method of Moments estimator.

Key distinction

When $\ell = k$, the sample moment equations can in principle be solved exactly, and the estimator is defined by setting the sample moments equal to zero. When $\ell > k$, there are more sample restrictions than unknown parameters, and this exact matching is typically no longer possible.

1.11 The Overidentified Case: GMM

We now turn to the more interesting case in which the model is *overidentified*:

$$\ell > k.$$

To fix ideas, consider the linear IV moment condition

$$g_i(\beta) = Z_i(Y_i - X_i'\beta),$$

where $Z_i \in \mathbb{R}^\ell$ and $\beta \in \mathbb{R}^k$. The sample moment is

$$\bar{g}_n(\beta) = \frac{1}{n} \sum_{i=1}^n Z_i(Y_i - X_i'\beta).$$

Using matrix notation, this can be written as

$$\bar{g}_n(\beta) = \frac{1}{n}(Z'Y - Z'X\beta).$$

In the overidentified case, there is generally no β such that

$$\bar{g}_n(\beta) = 0.$$

That is, the system typically has more equations than unknowns, so exact sample moment matching is not feasible.

The Generalized Method of Moments resolves this problem by choosing β so that the sample moments are *as close to zero as possible*, according to a weighted quadratic criterion.

Definition. Let W be a symmetric positive definite $\ell \times \ell$ matrix. The *GMM criterion function* is

$$J_n(\beta) = n \bar{g}_n(\beta)' W \bar{g}_n(\beta).$$

The *GMM estimator* is defined by

$$\hat{\beta}_{gmm} = \arg \min_{\beta \in B} J_n(\beta).$$

A proof of this can be found in Appendix 7.2.

Remark. The matrix W determines how different moment conditions are weighted in the criterion. Different choices of W lead to different GMM estimators in overidentified models.²

1.11.1 Linear IV GMM Estimator

For the linear IV model,

$$\bar{g}_n(\beta) = \frac{1}{n}(Z'Y - Z'X\beta),$$

so the criterion becomes

$$J_n(\beta) = n \left[\frac{1}{n}(Z'Y - Z'X\beta) \right]' W \left[\frac{1}{n}(Z'Y - Z'X\beta) \right].$$

It is convenient to define

$$A_n = \frac{1}{n}Z'X, \quad b_n = \frac{1}{n}Z'Y.$$

Then

$$\bar{g}_n(\beta) = b_n - A_n\beta,$$

²For now, the specific choice of W is not essential. Later, we will see that the choice of W affects the asymptotic variance of the GMM estimator. This means that W can be chosen to maximize asymptotic efficiency. At this stage, it is enough to assume that W is an $\ell \times \ell$ symmetric positive definite matrix, hence invertible.

and

$$J_n(\beta) = n(b_n - A_n\beta)'W(b_n - A_n\beta).$$

Differentiating with respect to β , the first-order condition is

$$0 = \frac{\partial}{\partial \beta} J_n(\hat{\beta}_{gmm}) = -2nA_n'W(b_n - A_n\hat{\beta}_{gmm}).$$

Substituting back for A_n and b_n ,

$$0 = -2n \left(\frac{1}{n} X'Z \right) W \left(\frac{1}{n} Z'(Y - X\hat{\beta}_{gmm}) \right).$$

Equivalently,

$$X'ZWZ'Y = X'ZWZ'X \hat{\beta}_{gmm}.$$

Therefore,

$$\hat{\beta}_{gmm} = (X'ZWZ'X)^{-1}(X'ZWZ'Y),$$

provided $X'ZWZ'X$ is invertible.

Proposition. *For the overidentified linear IV model, the GMM estimator has the explicit form*

$$\hat{\beta}_{gmm} = (X'ZWZ'X)^{-1}(X'ZWZ'Y).$$

1.11.2 Relation to 2SLS and Just-Identified IV

A particularly important special case is obtained by choosing

$$W = (Z'Z)^{-1}.$$

Then

$$\hat{\beta}_{gmm} = (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'Y,$$

which is exactly the two-stage least squares estimator:

$$\hat{\beta}_{gmm} = \hat{\beta}_{2sls}.$$

If instead the model is just-identified, so that $\ell = k$, then GMM collapses to IV. Indeed, if $Z'X$ is square and invertible, the first-order condition implies

$$Z'(Y - X\hat{\beta}) = 0,$$

and hence

$$\hat{\beta}_{gmm} = \hat{\beta}_{iv}.$$

Remark. In the just-identified case, the choice of weighting matrix does not affect the estimator,

because the sample moments can be driven exactly to zero. In contrast, in the overidentified case, the weighting matrix matters.

2 Distribution Theory and Efficient GMM

2.1 Asymptotic Distribution of the Linear GMM Estimator

We now study the asymptotic distribution of the GMM estimator in the linear IV model. Let

$$Y_i = X_i' \beta_0 + e_i,$$

and define the moment condition

$$g_i(\beta) = Z_i(Y_i - X_i' \beta).$$

At the true parameter β_0 , this becomes

$$g_i(\beta_0) = Z_i e_i.$$

Assume that a weighting matrix $\hat{W}$ is used such that

$$\hat{W} \xrightarrow{p} W,$$

where W is a fixed symmetric positive definite matrix. Also define

$$Q = \mathbb{E}[Z_i X_i'].$$

Under suitable regularity conditions, we have the law of large numbers result

$$\left(\frac{1}{n} X' Z \right) \hat{W} \left(\frac{1}{n} Z' X \right) \xrightarrow{p} Q' W Q.$$

Next, let $e = (e_1, \dots, e_n)'$. By a central limit theorem,

$$\frac{1}{\sqrt{n}} Z' e = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i e_i \xrightarrow{d} N(0, \Omega),$$

where

$$\Omega = \mathbb{E}[Z_i Z_i' e_i^2].$$

Therefore, by Slutsky's theorem,

$$\left(\frac{1}{n} X' Z \right) \hat{W} \left(\frac{1}{\sqrt{n}} Z' e \right) \xrightarrow{d} N(0, Q' W \Omega W Q).$$

To obtain the asymptotic distribution of $\hat{\beta}_{gmm}$, note that

$$\hat{\beta}_{gmm} = (X'Z\hat{W}Z'X)^{-1} (X'Z\hat{W}Z'Y).$$

Since $Y = X\beta_0 + e$,

$$\hat{\beta}_{gmm} - \beta_0 = (X'Z\hat{W}Z'X)^{-1} X'Z\hat{W}Z'e.$$

Multiplying by $\sqrt{n}$,

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta_0) = \left[\left(\frac{1}{n} X'Z \right) \hat{W} \left(\frac{1}{n} Z'X \right) \right]^{-1} \left(\frac{1}{n} X'Z \right) \hat{W} \left(\frac{1}{\sqrt{n}} Z'e \right).$$

Combining the previous convergence results yields

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta_0) \xrightarrow{d} N(0, V_\beta),$$

where

$$V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}.$$

Proposition. *Under suitable regularity conditions,*

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta_0) \xrightarrow{d} N(0, V_\beta),$$

with asymptotic variance

$$V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}, \quad \Omega = \mathbb{E}[Z_i Z_i' e_i^2].$$

Remark. This is the familiar sandwich form of the asymptotic variance. The outer terms come from the sensitivity of the moments to the parameter, while the middle term captures the asymptotic variability of the sample moments themselves.

2.2 Efficient GMM

The asymptotic variance depends on the weighting matrix W . This means that, in overidentified models, some choices of W lead to more efficient estimators than others.

The asymptotically efficient choice is

$$W = \Omega^{-1}, \quad \Omega = \mathbb{E}[Z_i Z_i' e_i^2].$$

Substituting this choice into the general variance formula gives

$$V_\beta^{eff} = (Q'\Omega^{-1}Q)^{-1}.$$

Definition. The *efficient GMM estimator* is the GMM estimator obtained by using the weighting

matrix

$$W = \Omega^{-1}.$$

Remark. Efficient GMM places relatively less weight on moment conditions that are noisy and relatively more weight on moment conditions that are precisely estimated. This is exactly what one would want from an efficient estimation procedure.

In practice, Ω is unknown and must be estimated. A standard approach is therefore *two-step GMM*:

1. Compute a preliminary consistent estimator $\tilde{\beta}$, for example using a simple weighting matrix.
2. Form residuals

$$\tilde{e}_i = Y_i - X_i' \tilde{\beta},$$

and estimate

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n Z_i Z_i' \tilde{e}_i^2.$$

3. Set

$$\hat{W} = \hat{\Omega}^{-1},$$

and compute the second-step GMM estimator.

Big picture

The just-identified case leads to Method of Moments: solve the sample moments exactly. The overidentified case leads to GMM: minimize a weighted quadratic form in the sample moments. In linear IV models, this produces a closed-form estimator, nests 2SLS as a special case, and has asymptotic variance

$$(Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}.$$

The efficient choice sets $W = \Omega^{-1}$, yielding the smallest asymptotic variance within the GMM class.

Remark. A natural question is: if identification problems arise because there are too few moment conditions, why not simply add more moments? The answer is that additional moment conditions cannot be invented arbitrarily. They must be valid implications of the model or of the underlying assumptions. Moreover, even valid extra moments need not improve identification if they are redundant or weakly informative. Thus, what matters is not only the number of moments, but also their validity and informational content.

2.3 Efficient GMM

The asymptotic variance of the linear GMM estimator is

$$V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1},$$

where

$$Q = \mathbb{E}[Z_i X_i'], \quad \Omega = \mathbb{E}[Z_i Z_i' e_i^2].$$

This expression makes clear that the choice of weighting matrix W affects the asymptotic behavior of $\hat{\beta}_{gmm}$. It is therefore natural to ask whether there is a choice of W that minimizes the asymptotic variance.

The answer is yes. The variance-minimizing weighting matrix is

$$W = \Omega^{-1}.$$

The corresponding estimator is called the *efficient GMM estimator*.

Definition. The *efficient GMM estimator* is the GMM estimator obtained by using the weighting matrix

$$W = \Omega^{-1}.$$

In the linear IV model, it takes the form

$$\hat{\beta}_{gmm}^{eff} = (X'Z\Omega^{-1}Z'X)^{-1}(X'Z\Omega^{-1}Z'Y).$$

Substituting $W = \Omega^{-1}$ into the general asymptotic variance formula yields

$$V_\beta^{eff} = (Q'\Omega^{-1}Q)^{-1}.$$

Proposition (Efficient weighting matrix). *For any symmetric positive definite matrix W , we have*

$$(Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1} - (Q'\Omega^{-1}Q)^{-1} \geq 0,$$

where the inequality is in the sense of positive semidefinite matrices.

Proof (Hayashi, 2000). Let

$$A = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}, \quad B = (Q'\Omega^{-1}Q)^{-1}.$$

Since both A and B are symmetric positive definite, a standard linear algebra result implies that

$$A - B \geq 0 \iff B^{-1} - A^{-1} \geq 0.$$

Therefore, it is enough to show that

$$D \equiv Q' \Omega^{-1} Q - (Q' W Q)(Q' W \Omega W Q)^{-1}(Q' W Q) \geq 0.$$

Because Ω^{-1} is symmetric positive definite, there exists a nonsingular matrix C such that

$$C' C = \Omega^{-1}.$$

Equivalently,

$$\Omega = C^{-1} C'^{-1}.$$

Now define

$$H \equiv C Q, \quad G \equiv C'^{-1} W Q.$$

Then

$$H' H = Q' C' C Q = Q' \Omega^{-1} Q,$$

$$G' G = Q' W C^{-1} C'^{-1} W Q = Q' W \Omega W Q,$$

and

$$H' G = Q' C' C'^{-1} W Q = Q' W Q.$$

Using these identities, we can rewrite D as

$$\begin{aligned} D &= Q' \Omega^{-1} Q - (Q' W Q)(Q' W \Omega W Q)^{-1}(Q' W Q) \\ &= Q' C' C Q - (Q' W Q)(Q' W C^{-1} C'^{-1} W Q)^{-1}(Q' W Q) \\ &= H' H - (H' G)(G' G)^{-1}(G' H) \\ &= H' \left[I - G(G' G)^{-1} G' \right] H. \end{aligned}$$

Let

$$M_G \equiv I - G(G' G)^{-1} G'.$$

This is the annihilator matrix associated with G . It is symmetric and idempotent:

$$M_G' = M_G, \quad M_G^2 = M_G.$$

Hence M_G is positive semidefinite. Therefore, for any conformable vector x ,

$$x' D x = x' H' M_G H x = (H x)' M_G (H x) \geq 0.$$

So D is positive semidefinite.

We have thus shown that

$$B^{-1} - A^{-1} \geq 0.$$

Applying again the inverse-ordering result for positive definite matrices, it follows that

$$A - B \geq 0,$$

that is,

$$(Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1} - (Q'\Omega^{-1}Q)^{-1} \geq 0.$$

This proves the claim. $\square$

Remark. The argument shows that efficient GMM is obtained by choosing the weighting matrix $W = \Omega^{-1}$. In that case, the asymptotic variance is minimal in the positive semidefinite ordering, so no other positive definite weighting matrix can do better asymptotically.

Remark. Efficient GMM gives relatively more weight to moment conditions that are precisely estimated and relatively less weight to moment conditions that are noisy. This is the same basic logic that underlies weighted least squares.

2.4 Estimating the Inverse Covariance Matrix

Efficient GMM is infeasible in population form because Ω is unknown. To implement it, we must estimate Ω , and then use the inverse of that estimate as the weighting matrix.

In the linear IV model, let $\tilde{\beta}$ be a preliminary consistent estimator, for example the 2SLS estimator, and define residuals

$$\tilde{e}_i = Y_i - X_i'\tilde{\beta}.$$

Using these residuals, define the estimated moment vector

$$\tilde{g}_i = Z_i\tilde{e}_i, \quad \bar{g}_n = \frac{1}{n} \sum_{i=1}^n \tilde{g}_i.$$

Two natural estimators of Ω are the *uncentered* and *centered* estimators:

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \tilde{g}_i \tilde{g}_i',$$

and

$$\hat{\Omega}^* = \frac{1}{n} \sum_{i=1}^n (\tilde{g}_i - \bar{g}_n)(\tilde{g}_i - \bar{g}_n)'$$

Remark. The uncentered estimator targets $\mathbb{E}[\tilde{g}_i \tilde{g}_i']$, while the centered estimator targets $\mathbf{Var}(\tilde{g}_i)$. Under correct specification, where $\mathbb{E}[g_i(\beta_0)] = 0$, both coincide asymptotically. However, the centered estimator is more robust because it still estimates the variance of the moments even when the sample moments are not centered exactly at zero.

When $\mathbb{E}[Z_i e_i] = 0$, both estimators are consistent for Ω . But $\hat{\Omega}^*$ is often preferred because

it estimates

$$\mathbf{Var}(Z_i e_i)$$

even in settings where one may worry that $E[Z_i e_i] \neq 0$ in finite samples or under misspecification.

Using either estimator, one obtains a feasible efficient GMM estimator by setting

$$\hat{W} = \hat{\Omega}^{-1} \quad \text{or} \quad \hat{W} = (\hat{\Omega}^*)^{-1}.$$

Definition. The *two-step GMM estimator* is obtained as follows:

1. Choose an initial symmetric positive definite weighting matrix W_0 , such as $W_0 = I_\ell$, and compute a preliminary consistent estimator $\tilde{\beta}$.
2. Form residuals $\tilde{e}_i = Y_i - X_i' \tilde{\beta}$ and estimate Ω .
3. Set the weighting matrix equal to the inverse of the estimated Ω , and recompute the GMM estimator.

Under suitable regularity conditions, the feasible efficient GMM estimator satisfies

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta_0) \xrightarrow{d} N(0, V_\beta), \quad V_\beta = (Q' \Omega^{-1} Q)^{-1}.$$

Remark. The preliminary estimator need not be efficient. Its role is only to provide residuals that allow a consistent estimate of Ω . Once Ω is estimated consistently, the second-step GMM estimator attains the efficient asymptotic variance.

2.5 Why 2SLS Is Efficient GMM under Homoskedasticity

Recall that 2SLS can be written as a GMM estimator with weighting matrix

$$\hat{W} = (Z'Z)^{-1},$$

or equivalently

$$\hat{W} = \left(\frac{1}{n} Z'Z \right)^{-1},$$

since multiplying the weighting matrix by a positive scalar does not change the minimizer of the GMM objective.

Remark. GMM is invariant to rescaling of the weighting matrix. If $c > 0$, then

$$\arg \min_{\beta} \bar{g}_n(\beta)' W \bar{g}_n(\beta) = \arg \min_{\beta} \bar{g}_n(\beta)' (cW) \bar{g}_n(\beta).$$

Thus only the relative weighting of the moments matters, not the overall scale of W .

By the weak law of large numbers and the continuous mapping theorem,

$$\left(\frac{1}{n}Z'Z\right)^{-1} \xrightarrow{p} \mathbb{E}[Z_i Z_i']^{-1}.$$

Therefore, in general, 2SLS does *not* coincide with efficient GMM, because efficient GMM uses

$$W = \Omega^{-1} = \mathbb{E}[Z_i Z_i' e_i^2]^{-1},$$

whereas 2SLS asymptotically uses a weighting matrix proportional to

$$\mathbb{E}[Z_i Z_i']^{-1}.$$

However, suppose conditional homoskedasticity holds:

$$\mathbb{E}[e_i^2 \mid Z_i] = \sigma^2.$$

Then

$$\Omega = \mathbb{E}[Z_i Z_i' e_i^2] = \mathbb{E}[Z_i Z_i' \mathbb{E}[e_i^2 \mid Z_i]] = \sigma^2 \mathbb{E}[Z_i Z_i'].$$

Hence

$$\Omega^{-1} = \frac{1}{\sigma^2} \mathbb{E}[Z_i Z_i']^{-1}.$$

Since GMM is invariant to multiplication of W by a positive scalar, the efficient weighting matrix is proportional to $\mathbb{E}[Z_i Z_i']^{-1}$, which is exactly the asymptotic weighting used by 2SLS.

Therefore, under conditional homoskedasticity,

$$\hat{\beta}_{2sls}$$

is asymptotically equivalent to efficient GMM.

Proposition. *Under conditional homoskedasticity,*

$$\mathbb{E}[e_i^2 \mid Z_i] = \sigma^2,$$

the 2SLS estimator is the efficient GMM estimator. In general, without homoskedasticity, 2SLS is not efficient.

Remark (Why proportionality is enough). In linear IV models, the GMM moments are

$$g_i(\beta) = Z_i(Y_i - X_i'\beta) = Z_i e_i.$$

Therefore, the covariance matrix of the moments is

$$\Omega = \mathbb{E}[g_i(\beta_0)g_i(\beta_0)'] = \mathbb{E}[Z_i Z_i' e_i^2].$$

Efficient GMM weights the moments by Ω^{-1} . Under conditional homoskedasticity,

$$\mathbb{E}[e_i^2 \mid Z_i] = \sigma^2.$$

Hence,

$$\Omega = \mathbb{E}[Z_i Z_i' \mathbb{E}[e_i^2 \mid Z_i]] = \sigma^2 \mathbb{E}[Z_i Z_i'].$$

Therefore,

$$\Omega^{-1} = \frac{1}{\sigma^2} \mathbb{E}[Z_i Z_i']^{-1}.$$

Thus the efficient weighting matrix is proportional to

$$\mathbb{E}[Z_i Z_i']^{-1}.$$

The scalar $1/\sigma^2$ is irrelevant for the GMM minimizer because multiplying the weighting matrix by a positive scalar does not change the objective's argmin. Since 2SLS uses the sample analog

$$\left(\frac{1}{n} Z' Z \right)^{-1},$$

2SLS is asymptotically equivalent to efficient GMM under homoskedasticity.

Main takeaway

In overidentified models, the choice of weighting matrix matters. The asymptotically optimal choice is

$$W = \Omega^{-1}, \quad \Omega = \mathbb{E}[Z_i Z_i' e_i^2].$$

Because Ω is unknown, efficient GMM is implemented using a feasible two-step procedure. Under conditional homoskedasticity, Ω is proportional to $\mathbb{E}[Z_i Z_i']$, so 2SLS coincides with efficient GMM. Without homoskedasticity, it generally does not.

2.6 Basic Inference

For statistical inference based on GMM, we need a consistent estimator of the asymptotic variance

$$V_\beta = (Q' W Q)^{-1} (Q' W \Omega W Q) (Q' W Q)^{-1},$$

where

$$Q = \mathbb{E}[Z_i X_i'], \quad \Omega = \mathbb{E}[Z_i Z_i' e_i^2].$$

A natural plug-in estimator is obtained by replacing population objects with their sample analogues. Let

$$\hat{Q} = \frac{1}{n} \sum_{i=1}^n Z_i X_i' = \frac{1}{n} Z' X,$$

and let $\hat{e}_i = Y_i - X_i' \hat{\beta}_{gmm}$. Using either the centered or uncentered estimator of Ω , define $\hat{\Omega}$. Then a natural estimator of V_β is

$$\hat{V}_\beta = (\hat{Q}' \hat{W} \hat{Q})^{-1} (\hat{Q}' \hat{W} \hat{\Omega} \hat{W} \hat{Q}) (\hat{Q}' \hat{W} \hat{Q})^{-1}.$$

In the special case of efficient two-step GMM, where $\hat{W} = \hat{\Omega}^{-1}$, this simplifies to

$$\hat{V}_\beta^{eff} = (\hat{Q}' \hat{\Omega}^{-1} \hat{Q})^{-1}.$$

Remark. This is the sample analogue of the efficient asymptotic variance

$$V_\beta^{eff} = (Q' \Omega^{-1} Q)^{-1}.$$

The simplification occurs because the efficient weighting matrix cancels the middle Ω term in the sandwich formula.

Under standard regularity conditions,

$$\hat{V}_\beta \xrightarrow{p} V_\beta.$$

Therefore, inference on smooth functions of the parameter can be based on the usual large-sample tools.

2.6.1 t-tests and Wald tests

Let $\theta = r(\beta)$ be a scalar parameter of interest, where $r(\cdot)$ is differentiable. Then, by the delta method,

$$\sqrt{n}(r(\hat{\beta}_{gmm}) - r(\beta_0)) \xrightarrow{d} N(0, R V_\beta R'),$$

where

$$R = \left. \frac{\partial r(\beta)}{\partial \beta'} \right|_{\beta=\beta_0}.$$

A consistent estimator of the asymptotic variance is

$$\widehat{\mathbf{Var}}(r(\hat{\beta}_{gmm})) = \frac{1}{n} \hat{R} \hat{V}_\beta \hat{R}',$$

where

$$\hat{R} = \left. \frac{\partial r(\beta)}{\partial \beta'} \right|_{\beta=\hat{\beta}_{gmm}}.$$

Thus asymptotically valid t -tests can be constructed in the usual way.

More generally, for testing q smooth restrictions

$$H_0 : r(\beta_0) = 0,$$

with $r(\beta) \in \mathbb{R}^q$, the Wald statistic is

$$W_n = n r(\hat{\beta}_{gmm})' (\hat{R} \hat{V}_\beta \hat{R}')^{-1} r(\hat{\beta}_{gmm}),$$

and under H_0 ,

$$W_n \xrightarrow{d} \chi_q^2.$$

Remark. So once a consistent variance estimator is available, inference for GMM proceeds just as in other asymptotically normal estimation problems: standard errors, t -tests, confidence intervals, and Wald tests all follow from the same large-sample approximation.

2.7 Overidentification Tests

An important feature of overidentified models is that they are testable. When $\ell > k$, the model imposes more moment restrictions than are needed to estimate the parameter vector. Under correct specification, all valid moment conditions should be approximately satisfied at the estimated parameter value.

This suggests testing whether the sample moments evaluated at $\hat{\beta}_{gmm}$ are “too far” from zero.

Remark. A loose but useful intuition is the following. Under overidentification, there are multiple pieces of information about β_0 . If the model is correctly specified, these different sources of information should be mutually consistent. If they are not, the overidentifying restrictions may fail.

Let

$$\bar{g}_n(\hat{\beta}_{gmm}) = \frac{1}{n} \sum_{i=1}^n g_i(\hat{\beta}_{gmm}).$$

The standard GMM overidentification test statistic is

$$J = J_n(\hat{\beta}_{gmm}) = n \bar{g}_n(\hat{\beta}_{gmm})' \hat{\Omega}^{-1} \bar{g}_n(\hat{\beta}_{gmm}),$$

where $\hat{\Omega}$ is a consistent estimator of the covariance matrix of the moments.

Definition. The J -test for overidentifying restrictions is based on the statistic

$$J = n \bar{g}_n(\hat{\beta}_{gmm})' \hat{\Omega}^{-1} \bar{g}_n(\hat{\beta}_{gmm}).$$

Under the null hypothesis that all moment conditions are correctly specified,

$$H_0 : \mathbb{E}[g_i(\beta_0)] = 0,$$

and under standard regularity conditions,

$$J \xrightarrow{d} \chi^2_{\ell-k}.$$

Remark. The degrees of freedom are $\ell - k$ because there are ℓ moment restrictions but only k free parameters. Intuitively, k of the moments are used to estimate the parameters, while the remaining $\ell - k$ moments provide overidentifying content that can be tested.

The decision rule is therefore:

$$\text{Reject } H_0 \quad \text{for large values of } J.$$

Remark. The J -test does not tell us which moment condition is invalid. It only tells us that the full set of overidentifying restrictions is collectively inconsistent with the data.

2.8 Endogeneity Tests

We now consider a classical application of GMM testing ideas: testing whether a regressor is endogenous.

Suppose the model is

$$Y_1 = Z_1' \beta_1 + Y_2' \beta_2 + e,$$

and assume that

$$\mathbb{E}[Z_1 e] = 0, \quad \mathbb{E}[Z_2 e] = 0.$$

Here Y_2 is a regressor whose exogeneity we want to test. The null and alternative hypotheses are

$$H_0 : \mathbb{E}[Y_2 e] = 0 \quad \text{versus} \quad H_1 : \mathbb{E}[Y_2 e] \neq 0.$$

Under the null, Y_2 is exogenous and can itself be used as an instrument. Under the alternative, it cannot. This suggests comparing two sets of moment conditions:

- the smaller set of moments, based on instruments

$$Z = (Z_1, Z_2),$$

which is valid under both H_0 and H_1 ;

- the larger set of moments, based on instruments

$$Z = (Z_1, Z_2, Y_2),$$

which is valid only under H_0 .

Let

$$J_{\text{small}}$$

denote the minimized efficient GMM criterion using the smaller instrument set, and let

$$J_{\text{large}}$$

denote the minimized efficient GMM criterion using the larger instrument set. The C -test statistic is

$$C = J_{\text{large}} - J_{\text{small}}.$$

Definition. The C -test (or difference-in- J test) for endogeneity is based on the increase in the minimized GMM criterion generated by imposing the additional orthogonality condition implied by exogeneity:

$$C = J_{\text{large}} - J_{\text{small}}.$$

Under H_0 , the additional moment conditions generated by treating Y_2 as exogenous are valid. Under standard regularity conditions,

$$C \xrightarrow{d} \chi_{k_2}^2,$$

where k_2 is the dimension of Y_2 .

Thus the decision rule is:

$$\text{Reject } H_0 \quad \text{if } C > c_{1-\alpha},$$

where $c_{1-\alpha}$ is the $(1 - \alpha)$ -quantile of the $\chi_{k_2}^2$ distribution.

Remark. The intuition is that the smaller set of moments is valid whether or not Y_2 is endogenous. The additional moments involving Y_2 , however, are valid only if Y_2 is exogenous. If adding these moments creates a large increase in the minimized GMM criterion, then the data reject the extra orthogonality conditions implied by exogeneity.

Remark. This test is closely related to the familiar Durbin–Wu–Hausman logic: compare an estimator that remains valid under endogeneity with one that is efficient under exogeneity. In the GMM framework, this comparison is naturally expressed as a difference in minimized criterion values.

Inference and specification testing in GMM

Once a consistent estimator of the asymptotic variance is available, standard large-sample inference for GMM follows from asymptotic normality. In overidentified models, the J -test evaluates whether the full set of moment restrictions is jointly compatible with the data. Endogeneity tests can then be formulated as difference-in- J tests that assess whether additional orthogonality conditions implied by exogeneity are supported by the data.

3 Nonlinear and Conditional Moment Equation Models

So far, we have mostly focused on linear models with unconditional moment restrictions. This framework is already broad and very useful, but many econometric applications require a more general setup. In particular, economic models are often nonlinear in the parameter of interest, and many identifying restrictions arise naturally as conditional moment restrictions rather than unconditional ones.

For that reason, it is useful to extend the GMM framework in two directions. First, we allow the moment function to be nonlinear in the parameter. Second, we consider models in which the population restriction is conditional on an information set. We begin with the nonlinear case.

3.1 GMM for Nonlinear Models

Suppose the parameter of interest is $\beta \in \mathbb{R}^k$, and let $g_i(\beta)$ be an $m \times 1$ vector of moment functions, possibly nonlinear in β . The population moment restriction is

$$\mathbb{E}[g_i(\beta_0)] = 0,$$

where β_0 denotes the true parameter value.

The sample analogue of the moment condition is

$$\bar{g}_n(\beta) = \frac{1}{n} \sum_{i=1}^n g_i(\beta).$$

Given a symmetric positive definite weighting matrix W , the nonlinear GMM estimator is defined as

$$\hat{\beta}_{\text{gmm}} = \arg \min_{\beta} J_n(\beta), \quad J_n(\beta) = n \bar{g}_n(\beta)' W \bar{g}_n(\beta).$$

Remark. This is exactly the same logic as in linear GMM. The only difference is that the moment function is now nonlinear in β , so the minimization problem is generally solved numerically rather than in closed form.

To describe the asymptotic distribution of $\hat{\beta}_{\text{gmm}}$, define

$$Q = \mathbb{E} \left[\frac{\partial}{\partial \beta'} g_i(\beta_0) \right], \quad \Omega = \mathbb{E}[g_i(\beta_0) g_i(\beta_0)'].$$

Here Q is the Jacobian matrix of the population moment function, evaluated at the truth, and Ω is the covariance matrix of the moment conditions.

Under standard regularity conditions,

$$\sqrt{n}(\hat{\beta}_{\text{gmm}} - \beta_0) \xrightarrow{d} N(0, V_{\beta}),$$

where

$$V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}.$$

Remark. This is the nonlinear analogue of the usual GMM asymptotic variance formula. The structure is the same as before: the Jacobian Q measures how informative the moments are about the parameter, while Ω captures the sampling variability of the moments themselves.

If the efficient weighting matrix is used, namely

$$W = \Omega^{-1},$$

then the asymptotic variance simplifies to

$$V_\beta = (Q'\Omega^{-1}Q)^{-1}.$$

Remark. Thus, the same efficiency logic carries over from the linear case to the nonlinear one. Efficient GMM reweights the moments using the inverse of their covariance matrix, placing more weight on moments that are more informative and less noisy.

In practice, V_β is estimated by replacing the population objects Q and Ω with consistent sample analogues. This yields a feasible estimator of the asymptotic variance and therefore standard errors for nonlinear GMM estimation.

Nonlinear GMM

Nonlinear GMM extends the usual moment-based framework by allowing $g_i(\beta)$ to be nonlinear in β . The estimator still minimizes a quadratic form in the sample moments, and its large-sample distribution has the same sandwich form as in linear GMM. The efficient choice of weighting matrix remains $W = \Omega^{-1}$.

3.2 Conditional Moment Equation Models

Up to this point, we have mainly worked with *unconditional* moment restrictions of the form

$$\mathbb{E}[g_i(\beta_0)] = 0.$$

In many econometric applications, however, the model implies a stronger restriction: not merely that a residual has mean zero on average, but that it has mean zero *conditional* on a vector of observable variables Z_i . This leads to the conditional moment restriction

$$\mathbb{E}[e_i(\beta_0) \mid Z_i] = 0.$$

A useful way to think about this condition is the following. An unconditional moment restriction says that the error averages out in the population. A conditional moment restriction

says something stronger: once we condition on the information contained in Z_i , the error is still not systematically predictable. In that sense, conditional moments contain much more information than unconditional ones.

A leading example is the regression equation

$$Y_i = X_i' \beta_0 + e_i,$$

together with the restriction

$$\mathbb{E}[e_i \mid Z_i] = 0.$$

Equivalently,

$$\mathbb{E}[Y_i - X_i' \beta_0 \mid Z_i] = 0.$$

Thus, the identifying content of the model is naturally expressed as a conditional moment equation.

Proposition (From conditional to unconditional moments). *Suppose*

$$\mathbb{E}[e_i(\beta_0) \mid Z_i] = 0.$$

Then, for any measurable function $\phi(Z_i, \beta_0)$ such that the expectation exists,

$$\mathbb{E}[\phi(Z_i, \beta_0)e_i(\beta_0)] = 0.$$

Proof. By the law of iterated expectations,

$$\mathbb{E}[\phi(Z_i, \beta_0)e_i(\beta_0)] = \mathbb{E}[\mathbb{E}[\phi(Z_i, \beta_0)e_i(\beta_0) \mid Z_i]].$$

Since $\phi(Z_i, \beta_0)$ is measurable with respect to Z_i ,

$$\mathbb{E}[\phi(Z_i, \beta_0)e_i(\beta_0) \mid Z_i] = \phi(Z_i, \beta_0)\mathbb{E}[e_i(\beta_0) \mid Z_i] = 0.$$

Therefore,

$$\mathbb{E}[\phi(Z_i, \beta_0)e_i(\beta_0)] = 0. \quad \square$$

This proposition is fundamental. It shows that a single conditional moment restriction generates an entire class of valid unconditional moment restrictions. For every admissible function $\phi(Z_i, \beta)$, we obtain a moment condition that can in principle be used in GMM.

Remark. This is where the main new issue arises. In unconditional GMM, we typically start with a finite list of moments. In conditional moment models, by contrast, there are potentially infinitely many valid moments, because there are infinitely many functions of Z_i . The question is therefore not only whether valid moments exist, but which instrument functions should be used.

This leads to three natural approaches:

1. choose a finite set of instrument functions from some rich class;
2. construct the *optimal* instrument, meaning the one that minimizes the asymptotic variance;
3. use a simple instrument choice, such as Z_i itself, and proceed with standard IV or 2SLS.

The first approach is conceptually attractive but technically delicate, because one must decide how to approximate an infinite-dimensional instrument space with a finite-dimensional class. The third approach is simple and often practical, but it need not be efficient. The second approach is the one that delivers the efficiency benchmark.

Optimal instruments

The efficient choice of instrument in conditional moment models was derived by Chamberlain. Let

$$\sigma_i^2 = \mathbb{E}[e_i(\beta_0)^2 \mid Z_i].$$

Under regularity conditions, the optimal instrument is

$$A_i = -\sigma_i^{-2} \mathbb{E} \left[\frac{\partial}{\partial \beta} e_i(\beta_0) \mid Z_i \right],$$

so that the efficient moment condition is

$$g_i(\beta_0) = A_i e_i(\beta_0).$$

Remark. This formula has a very natural interpretation. The term

$$\mathbb{E} \left[\frac{\partial}{\partial \beta} e_i(\beta_0) \mid Z_i \right]$$

captures how sensitive the residual is to changes in the parameter, conditional on the information in Z_i . The factor σ_i^{-2} then reweights observations by their conditional precision, giving less weight where the conditional error variance is large and more weight where it is small. Thus, the optimal instrument balances *identification strength* and *noise reduction*.

Remark. This is the conditional-moment analogue of the efficient weighting logic in GMM and the precision-weighting logic in GLS. The estimator uses exactly those transformations of Z_i that are most informative about β_0 , while discounting components associated with high conditional variance.

Linear IV as a special case

Consider the linear model

$$e_i(\beta) = Y_i - X_i'\beta.$$

Then

$$\frac{\partial}{\partial \beta} e_i(\beta) = -X_i,$$

so the optimal instrument becomes

$$A_i = \sigma_i^{-2} \mathbb{E}[X_i | Z_i].$$

This is a very useful result. It shows that, in the linear IV model, the efficient instrument is the conditional expectation of the endogenous regressor given the instrument set, reweighted by conditional precision.

If, moreover, $X_i = Z_i$, then

$$A_i = \sigma_i^{-2} X_i.$$

In that case, efficient GMM coincides with GLS.

Remark. The linear formula makes the general result much easier to understand. One should use, from the information contained in Z_i , exactly the component that best predicts X_i , and then weight it more heavily when the conditional noise is small. This is the deeper efficiency logic underlying instrumental variables estimation.

In practice, the optimal instrument A_i is unknown, because it depends on conditional expectations and conditional variances that must themselves be estimated. For that reason, the optimal IV formula is primarily a benchmark: it tells us what the efficient estimator is trying to approximate.

Why conditional moments matter

A conditional moment restriction such as

$$\mathbb{E}[e_i(\beta_0) | Z_i] = 0$$

contains much more information than a single unconditional orthogonality condition. By multiplying the residual by functions of Z_i , we obtain many valid unconditional moments. The econometric problem is then to choose instrument functions that extract this information as efficiently as possible. Chamberlain's optimal IV result provides the efficiency benchmark for this choice.

3.3 Empirical Illustration: Revisiting Angrist and Krueger (1991) Using GMM

As an empirical illustration, consider the classic returns-to-schooling specification

$$\log(wage_i) = \beta_1 + \beta_2 education_i + X_i' \beta_3 + e_i.$$

We focus on the specification without state-of-birth indicators and use only the three quarter-of-birth indicators as instruments, so that $\ell_2 = 3$.

Table 1 reports the resulting estimates from efficient GMM and conventional 2SLS.

	(1) GMM	(2) 2SLS
$\hat{\beta}_2$	0.0990964	0.0989901
$se(\hat{\beta}_2)$	(0.0207334)	(0.0207315)
J -statistic	2.34278	
p -value	(0.3099)	
C-statistic	3.09299	
p -value	(0.0786)	

Table 1: Angrist and Krueger (1991) revisited with GMM

Several points are worth emphasizing.

First, the GMM and 2SLS estimates of the return to education are nearly identical:

$$\hat{\beta}_2 \approx 0.099.$$

Under the usual semi-log interpretation, this corresponds to an estimated return of roughly 9.9% in wages for an additional year of schooling.

Second, the standard errors are also essentially the same. This suggests that, in this specification, efficient reweighting through GMM yields only limited gains relative to conventional 2SLS. That is not surprising: when the number of instruments is small and the moment structure is relatively simple, 2SLS is often already close to the efficient estimator.

Third, because the model is overidentified, the J -test can be used to assess the joint validity of the overidentifying restrictions. The reported p -value of 0.3099 does not reject the null, so the data do not provide evidence against the validity of the instrument set.

Finally, the reported C-statistic equals 3.09299, with p -value 0.0786. Interpreted as a difference-in- J test, this provides only mild evidence against the null being tested: it would reject at the 10% level, but not at the 5% level.

Remark. This illustration is useful because it connects the abstract efficiency theory to a familiar empirical application. In practice, GMM often confirms the main IV estimate while also providing a unified framework for overidentification tests, endogeneity tests, and efficient weighting.

Empirical takeaway

In the Angrist–Krueger application, efficient GMM and 2SLS deliver almost identical estimates of the return to education. The overidentifying restrictions are not rejected, and the difference-in- J statistic suggests at most weak evidence against the null under consideration. This is a good example of how GMM extends IV without necessarily overturning its main empirical conclusions.

3.4 A useful intuition: GMM as a weighted average of just-identified IV estimators

Consider the scalar linear model

$$Y_i = X_i\beta + e_i,$$

and suppose we have access to two scalar instruments, Z_{1i} and Z_{2i} . A convenient way to think about GMM in this setting is that it tries to make the sample moments

$$\frac{1}{n} \sum_{i=1}^n Z_{1i}(Y_i - X_i\beta) \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n Z_{2i}(Y_i - X_i\beta)$$

as close to zero as possible, while allowing us to weight the two moment conditions differently.

Define

$$\bar{g}_n(\beta) = \begin{pmatrix} \frac{1}{n} \sum_{i=1}^n Z_{1i}(Y_i - X_i\beta) \\ \frac{1}{n} \sum_{i=1}^n Z_{2i}(Y_i - X_i\beta) \end{pmatrix}, \quad W = \begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_2 \end{pmatrix}, \quad \omega_1, \omega_2 > 0.$$

Then the GMM estimator solves

$$\hat{\beta}_{GMM} = \arg \min_{\beta} \bar{g}_n(\beta)' W \bar{g}_n(\beta).$$

Equivalently, one may write

$$\hat{\beta}_{GMM} = \arg \min_{\beta} n \bar{g}_n(\beta)' W \bar{g}_n(\beta),$$

since multiplying the objective function by the positive constant n does not change the minimizer. The factor n is useful for asymptotic theory and for the overidentification J -test, but it is irrelevant for finding the estimator itself.

To see the structure more clearly, define

$$a_j = \frac{1}{n} \sum_{i=1}^n Z_{ji}Y_i, \quad b_j = \frac{1}{n} \sum_{i=1}^n Z_{ji}X_i, \quad j = 1, 2.$$

Then

$$\bar{g}_n(\beta) = \begin{pmatrix} a_1 - b_1\beta \\ a_2 - b_2\beta \end{pmatrix},$$

so the criterion becomes

$$J_n(\beta) = \omega_1(a_1 - b_1\beta)^2 + \omega_2(a_2 - b_2\beta)^2.$$

Differentiating and solving the first-order condition yields

$$\hat{\beta}_{GMM} = \frac{\omega_1 b_1 a_1 + \omega_2 b_2 a_2}{\omega_1 b_1^2 + \omega_2 b_2^2}.$$

Now note that the just-identified IV estimator using only Z_1 is

$$\hat{\beta}_{IV,1} = \frac{a_1}{b_1},$$

and the one using only Z_2 is

$$\hat{\beta}_{IV,2} = \frac{a_2}{b_2}.$$

Substituting these expressions into the GMM formula gives

$$\hat{\beta}_{GMM} = \frac{\omega_1 b_1^2}{\omega_1 b_1^2 + \omega_2 b_2^2} \hat{\beta}_{IV,1} + \frac{\omega_2 b_2^2}{\omega_1 b_1^2 + \omega_2 b_2^2} \hat{\beta}_{IV,2}.$$

Therefore, $\hat{\beta}_{GMM}$ is a weighted average of the two IV estimators obtained separately from Z_1 and Z_2 .

Main intuition. This example is useful because it makes the logic of GMM very transparent. Each instrument generates one sample moment condition, and each moment condition by itself produces a just-identified IV estimator. GMM combines these moment conditions into a single criterion and chooses the value of β that makes the weighted sum of squared moment errors as small as possible.

The effective weight attached to each IV estimator is not only ω_j , but $\omega_j b_j^2$. The term ω_j reflects how much importance the weighting matrix gives to moment j , while b_j^2 captures how strongly that moment responds to changes in β . Thus, a moment condition receives more influence when it is both highly weighted and highly informative about the parameter.

In this sense, GMM in overidentified models can be viewed as a disciplined way of combining several just-identified estimators into a single estimator that uses all available moment information.

4 Potential Outcomes Framework

References

The main references for this section are Angrist and Pischke (2009) and Imbens and Rubin (2015). An additional useful reference is Angrist and Pischke (2015). For an empirical application, we will later refer to Finkelstein et al. (2012).

Motivating example: Do hospitals make you sick?

A useful motivating example, following Angrist and Pischke (2009), is the question:

Do hospitals make you sick?

Consider data from the 2005 NHIS. Let the outcome of interest be

$$Y_i \in \{1, 2, 3, 4, 5\},$$

where Y_i is self-assessed health, with 1 denoting poor health and 5 denoting very good health.

Define the treatment indicator by

$$D_i = \begin{cases} 1, & \text{if individual } i \text{ had an overnight hospital stay during the last 12 months,} \\ 0, & \text{otherwise.} \end{cases}$$

The raw comparison between treated and untreated individuals is summarized in Table 2.

	N	$\bar{Y}$	SE
$D = 1$	7,774	3.21	0.014
$D = 0$	90,049	3.93	0.003

Table 2: Hospital stay and self-assessed health

The difference in average outcomes is

$$\bar{Y}_1 - \bar{Y}_0 = 3.21 - 3.93 = -0.72,$$

and this difference is statistically significant.

A first, naive reading of this comparison might suggest that hospital stays worsen health. But this conclusion is not yet justified as a causal statement. Individuals who go to the hospital are systematically different from those who do not, so a simple comparison of means generally mixes together causal effects and selection effects.

Remark. This is exactly the type of problem that motivates the potential outcomes framework. The central question is not merely whether treated and untreated units have different average outcomes, but whether the observed difference can be interpreted causally.

Introduction

So far, we have discussed causality mainly through linear models. To illustrate the limitations of that approach, recall the type of model used in earlier empirical work:

$$Y_2 = \alpha + \gamma Y_1 + \beta \textit{Treatment} + e_2,$$

where β is interpreted as the causal effect of interest.

When this is taken to be the true causal model, it imposes restrictive assumptions. In particular:

1. β is a constant, so treatment effects are homogeneous.
2. The model is linear and additive, so treatment effects do not vary with baseline outcomes or unobservables.

The potential outcomes framework provides a much more general language for defining and analyzing causal effects. It does not require causality to be embedded in a specific linear or nonlinear regression model. Instead, it starts from a conceptual description of what would happen to each unit under alternative treatment states.

Historically, this framework was introduced by Jerzy Neyman and later generalized and popularized by Donald Rubin.

Why move beyond linear causal models?

Linear causal models are often convenient, but they can build in strong restrictions from the start. The potential outcomes framework separates the definition of causal effects from the choice of statistical model. This is powerful because it lets us define causality first and only then ask what assumptions are needed to identify it from the data.

The potential outcomes framework

We now introduce the potential outcomes (PO) framework, which provides a general language for causal inference.

We are interested in the causal effect of a binary treatment, policy, or intervention

$$D_i \in \{0, 1\}$$

on an outcome

$$Y_i \in \mathcal{Y} \subseteq \mathbb{R}.$$

In addition, for each unit i , we observe a vector of covariates

$$X_i \in \mathcal{X} \subseteq \mathbb{R}^k.$$

The key conceptual step in the PO framework is to postulate the existence of two potential outcomes for each unit, indexed by treatment status:

$$Y_i(1) \quad \text{and} \quad Y_i(0),$$

where

- $Y_i(1)$ is the outcome that unit i would realize if treated ($D_i = 1$),
- $Y_i(0)$ is the outcome that unit i would realize if untreated ($D_i = 0$).

The individual causal effect is then defined as

$$Y_i(1) - Y_i(0).$$

Remark. This is an extremely important conceptual shift. In the PO framework, causality is defined at the level of *counterfactual comparisons*: for a given unit, the causal effect is the difference between what would happen under treatment and what would happen without treatment.

Remark. The individual treatment effect $Y_i(1) - Y_i(0)$ is itself a random variable. This immediately allows for heterogeneous treatment effects: different units may respond differently to the same intervention.

A useful intuition

The PO framework asks a simple but powerful question: for the same unit, what would the outcome be under treatment, and what would it be without treatment? Causal inference is fundamentally about comparing these two states of the world.

The fundamental challenge of causal inference

If both $Y_i(1)$ and $Y_i(0)$ were observed for every unit, then causal inference would be straightforward. We could compute the treatment effect for each unit,

$$Y_i(1) - Y_i(0),$$

and therefore recover the entire distribution of treatment effects.

The problem is that, in reality, we only observe one of the two potential outcomes for each unit:

- if $D_i = 1$, we observe $Y_i(1)$,
- if $D_i = 0$, we observe $Y_i(0)$.

We do *not* observe $Y_i(1)$ for units with $D_i = 0$, and we do *not* observe $Y_i(0)$ for units with $D_i = 1$.

Thus, the realized outcome can be written as

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i).$$

Equivalently,

$$Y_i = Y_i(0) + (Y_i(1) - Y_i(0))D_i.$$

These identities are purely algebraic, but they are very revealing. They show that the observed outcome combines two objects:

1. the untreated potential outcome $Y_i(0)$,
2. the treatment effect $Y_i(1) - Y_i(0)$, which is switched on only when $D_i = 1$.

Remark. This is the fundamental problem of causal inference: for each unit, one potential outcome is observed and the other is missing. The missing potential outcome is the counterfactual.

Remark. Because one of the two potential outcomes is always unobserved, causal effects cannot be identified from the data alone. Additional assumptions are needed to link observed outcomes to the counterfactual outcomes we do not see.

Why causality is hard

Data never show us both states of the world for the same unit. We see what happened under the realized treatment status, but we do not see what would have happened under the alternative treatment status. Causal inference is therefore a missing-data problem at its core.

Defining the objects of interest

In principle, once the potential outcomes $Y(1)$ and $Y(0)$ are introduced, one could study any feature of their joint distribution,

$$F_{Y(0),Y(1)}(y_0, y_1) = P(Y(0) \leq y_0, Y(1) \leq y_1).$$

This joint distribution fully characterizes how untreated and treated potential outcomes are related across units.

For example, one might be interested in the distribution of individual treatment effects,

$$F_{Y(1)-Y(0)}(y) = P(Y(1) - Y(0) \leq y).$$

Unfortunately, identifying either the joint distribution $F_{Y(0),Y(1)}$ or the distribution of treatment effects $F_{Y(1)-Y(0)}$ is generally impossible without very strong assumptions, because we never observe both potential outcomes for the same unit.

Therefore, following standard practice, we focus on causal parameters that are identifiable under weaker assumptions.

A useful benchmark is to ask what is identified in an experiment where treatment is randomly assigned. That benchmark will guide the parameters we emphasize below.

Remark. The key difficulty is that the marginal distributions of $Y(1)$ and $Y(0)$ may be identified under suitable assumptions, but their *joint* distribution is much harder to recover because the dependence between the two potential outcomes is fundamentally unobserved.

Remark. To identify the full distribution of treatment effects, one would need additional structure tying together $Y(1)$ and $Y(0)$, such as strong restrictions on their dependence. This is why most empirical work concentrates on averages, quantiles, or other functionals that can be learned from weaker assumptions.

Treatment effects

The main treatment effect considered in these notes is the Average Treatment Effect (ATE),

$$ATE = \mathbb{E}[Y(1) - Y(0)].$$

The ATE is the average effect that would arise if we could compare the entire population under treatment with that same population without treatment. This is the ideal causal comparison, but it is fundamentally counterfactual because each unit is observed in only one treatment state.

A second important object is the Average Treatment Effect on the Treated (ATT),

$$ATT = \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

The ATT focuses only on the units that actually received the treatment. It asks: among treated units, what is the average difference between their outcome under treatment and the outcome they would have had without treatment? The second object is counterfactual, but under additional assumptions it can sometimes be identified from the data.

Other treatment effects of interest include Distributional Treatment Effects (DTE), defined by

$$DTE(y) = F_{Y(1)}(y) - F_{Y(0)}(y), \quad y \in \mathcal{Y},$$

The DTE compares the entire distribution of potential outcomes under treatment with the entire distribution without treatment. Instead of asking how the mean changes, it asks how the probability of being below each outcome level y changes when moving from no treatment to treatment.

And Quantile Treatment Effects (QTE), defined by

$$QTE(\tau) = Q_{Y(1)}(\tau) - Q_{Y(0)}(\tau), \quad \tau \in (0, 1),$$

where $Q_{Y(d)}(\tau)$ denotes the τ -quantile of the distribution of $Y(d)$, for $d \in \{0, 1\}$.

The QTE compares quantiles of the treated and untreated potential outcome distributions. It asks how the treatment changes different parts of the outcome distribution, such as the median or the lower and upper tails. In general, this is a comparison of marginal quantiles, not necessarily the treatment effect for the same individual located at a given quantile.

One may also define the corresponding objects for the treated subpopulation, such as treated-group distributional effects and treated-group quantile treatment effects.

Remark. The ATE is a mean comparison across the two potential outcome distributions. The ATT instead focuses on the subpopulation that actually receives treatment. These two parameters coincide only under special conditions.

Remark. DTE and QTE are useful when treatment effects are heterogeneous. The ATE can mask important distributional changes, whereas DTE and QTE allow us to see whether treatment shifts the entire outcome distribution or affects some parts of it more than others.

A useful hierarchy of causal parameters

The potential outcomes framework naturally generates a hierarchy of causal objects. At the most ambitious level, one may want the full joint distribution of $(Y(0), Y(1))$. In practice, this is rarely identified. More commonly, empirical work targets lower-dimensional summaries such as the ATE, ATT, DTE, or QTE, which require weaker assumptions and are often directly policy-relevant.

SUTVA

An important assumption that is often left implicit in the potential outcomes framework is the Stable Unit Treatment Value Assumption (SUTVA).

Proposition (SUTVA). *The Stable Unit Treatment Value Assumption has two components:*

1. *The potential outcomes for any unit do not vary with the treatments assigned to other units.*
2. *For each unit, there are not different hidden versions of a given treatment status that generate different potential outcomes.*

The first component rules out *interference* across units. In other words, one unit's outcome cannot be affected by whether other units are treated. This excludes settings with peer effects, spillovers, equilibrium effects, contagion, and related interactions.

The second component rules out *hidden variation in treatment*. The label $D = 1$ must correspond to a well-defined intervention, and similarly for $D = 0$. If different versions of treatment produce different outcomes, then the notation $Y(1)$ is not sufficiently precise.

Under SUTVA, it is meaningful to speak of exactly two potential outcomes for each unit, $Y_i(1)$ and $Y_i(0)$, indexed only by that unit's own treatment status.

Remark. SUTVA is conceptually basic but substantively strong. It is often plausible in tightly controlled experiments, but it can easily fail in economic settings where agents interact, markets adjust, or treatments are implemented in heterogeneous ways.

Remark. Without SUTVA, the notation $Y_i(1)$ and $Y_i(0)$ may be misleading. With interference, one would need potential outcomes indexed by the entire treatment assignment vector. With multiple versions of treatment, one would need a more refined treatment space than the binary variable D .

Why SUTVA matters

The potential outcomes framework starts by assigning to each unit two counterfactual outcomes, one under treatment and one under no treatment. SUTVA is what makes this two-outcome representation coherent. If other units' treatments matter, or if treatment has multiple hidden versions, then the simple binary potential-outcomes notation is no longer adequate.

Selection bias and random assignment

Simple mean comparisons are not causal effects

Consider the regression of Y on a constant and the treatment indicator D . Under standard regularity conditions, the OLS coefficient on D , say $\hat{\beta}$, is consistent for the difference in population means:

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0 \xrightarrow{P} \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

A natural question is whether

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$$

can be interpreted as a causal effect.

In general, the answer is no. Using the potential-outcomes representation

$$Y = DY(1) + (1 - D)Y(0),$$

we obtain

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0].$$

Add and subtract $\mathbb{E}[Y(0) \mid D = 1]$ to get

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid D = 1]}_{ATT} + \underbrace{\left(\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \right)}_{\text{selection bias}}.$$

Thus, the simple difference in observed means is equal to the Average Treatment Effect on the Treated only if the selection bias term is zero.

Remark. This decomposition is fundamental. It shows that observed differences between treated and untreated units combine two forces: a genuine causal effect and a pre-existing difference between the groups.

Remark. If treatment status D is related to the untreated potential outcome $Y(0)$, then the selection bias term is generally nonzero. In that case, comparing treated and untreated units confounds treatment effects with selection.

Intuition for selection bias

Selection bias arises because the treated and untreated groups are typically not comparable before treatment. The treated units may already differ from the untreated units in ways that matter for the outcome.

In the hospital example, people who spend the night in a hospital are usually in worse health to begin with. So when we compare their observed health to the health of people who did not go to the hospital, the difference reflects not only the effect of the hospital stay, but also the fact that the hospital group had worse baseline health even in the absence of treatment.

Formally, the term

$$\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]$$

measures exactly this baseline difference: how treated and untreated units would compare if neither group were treated. If this term is not zero, then a raw comparison of means cannot be given a causal interpretation.

This discussion suggests that one route to identifying causal effects is to search for settings in which the selection bias term is equal to zero.

Random assignment in experiments

Experiments provide a natural benchmark for observational research and are often described as the gold standard for causal inference.

The key reason is that researchers control the treatment assignment mechanism and assign treatment randomly. In a completely randomized experiment,

$$(Y(1), Y(0), X) \perp D.$$

That is, treatment assignment is statistically independent of both potential outcomes and observed covariates.

Under random assignment, treated and untreated units are comparable in expectation. In

particular, the untreated potential outcomes of treated units have the same distribution as the untreated potential outcomes of untreated units, and similarly for treated potential outcomes.

Remark. Random assignment eliminates selection bias because treatment status is no longer chosen on the basis of characteristics related to potential outcomes. The treated group is just a random subset of the population, and so is the control group.

Remark. There are many forms of experimental assignment. Here the focus is on complete or unconditional randomization, where treatment is assigned independently of everything else, for example by a random number generator.

What is identified under random assignment?

Under random assignment, the marginal distributions of both potential outcomes are identified. For each $d \in \{0, 1\}$,

$$F_{Y(d)}(y) = P(Y(d) \leq y).$$

Because $Y(d) \perp D$, we have

$$P(Y(d) \leq y) = P(Y(d) \leq y \mid D = d).$$

But when $D = d$, the observed outcome satisfies $Y = Y(d)$. Therefore,

$$F_{Y(d)}(y) = P(Y(d) \leq y \mid D = d) = P(Y \leq y \mid D = d).$$

Hence, the distribution of observed outcomes in the treated group identifies the distribution of $Y(1)$, and the distribution of observed outcomes in the control group identifies the distribution of $Y(0)$.

Similarly, for $d \in \{0, 1\}$, the treated-subpopulation distributions

$$F_{Y(d)|D=1}(y)$$

are identified under random assignment as well.

It follows that any functional of these marginal distributions is identified. In particular, this includes the ATE, ATT, DTE, and QTE.

Remark. Random assignment identifies the *marginal* distributions of $Y(1)$ and $Y(0)$, but not their joint distribution. Even in an experiment, we still do not observe both potential outcomes for the same unit.

Remark. This distinction is crucial. From the experiment, we can learn how outcomes are distributed under universal treatment and under universal non-treatment. But we still cannot directly recover the distribution of the individual treatment effect $Y(1) - Y(0)$, because that object depends on how $Y(1)$ and $Y(0)$ are paired within the same unit.

Intuition for the marginal distributions

The word “marginal” here means that we study the distribution of each potential outcome separately.

Under random assignment, the treated units give us a random sample from the population under treatment, so their observed outcomes reveal the distribution of $Y(1)$. Likewise, the untreated units give us a random sample from the population under no treatment, so their observed outcomes reveal the distribution of $Y(0)$.

This is why an experiment lets us identify objects such as

$$\mathbb{E}[Y(1)], \quad \mathbb{E}[Y(0)], \quad F_{Y(1)}(y), \quad F_{Y(0)}(y),$$

and therefore also the ATE, DTE, and QTE.

But the experiment still does not show us, for any one individual, both what happened under treatment and what would have happened without treatment. So it does not identify the joint distribution of $(Y(0), Y(1))$, nor the full distribution of individual treatment effects, without additional assumptions.

Discussion and intuition

The reason the ATT is identified under random assignment is that random assignment eliminates the selection bias term. Recall that

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = ATT + \left(\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \right).$$

Under random assignment,

$$Y(0) \perp D,$$

so

$$\mathbb{E}[Y(0) \mid D = 1] = \mathbb{E}[Y(0) \mid D = 0].$$

Therefore, the selection bias term is zero, and

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 1] = ATT.$$

Moreover, under random assignment we also have

$$Y(1) \perp D \quad \text{and} \quad Y(0) \perp D.$$

Hence,

$$ATT = \mathbb{E}[Y(1) - Y(0) \mid D = 1] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = ATE.$$

Remark. This is the key benchmark result. In a randomized experiment, the treated group

is not a selected group with systematically different counterfactual outcomes. It is simply a random subset of the population. Therefore, the untreated outcomes of the control group provide a valid estimate of the missing counterfactual outcomes for the treated group.

Remark. The equality $ATT = ATE$ under random assignment is also intuitive. Since treatment is assigned independently of potential outcomes, the treated units are representative of the full population in expectation. So the average effect for treated units is the same as the average effect in the population.

Intuition: why random assignment solves the selection problem

The problem in observational comparisons is that treated and untreated units are usually different *before* treatment. Random assignment breaks that link. It makes treatment status unrelated, in expectation, to what units would have looked like without treatment and also to what they would look like with treatment.

As a result, the control group becomes a valid stand-in for the missing counterfactual of the treated group. This is why the difference in observed means has a causal interpretation in a randomized experiment.

The Oregon Health Insurance Experiment

A leading empirical example is the Oregon Health Insurance Experiment, discussed in Finkelstein et al. (2012). In 2008, a group of uninsured low-income adults in Oregon was selected by lottery and given the chance to apply for Medicaid. This created a randomized controlled design that can be used to study the effects of expanded access to public health insurance.

As summarized in the abstract of Finkelstein et al. (2012), the Oregon lottery provides an opportunity to measure the effects of expanding public health insurance on health care use, financial strain, and health outcomes among low-income adults.

For details and additional publications, see

<https://www.nber.org/oregon>.

This example is especially useful because it allows us to connect the abstract potential-outcomes framework to the practical analysis of randomized experiments using regression methods.

Remark. A subtle but important feature of the Oregon design is that the lottery randomizes the *opportunity to apply* for Medicaid, not necessarily final insurance coverage itself. This distinction becomes important later when one studies imperfect compliance and instrumental variables.

Selection into health insurance: descriptive evidence

To see why selection bias is such a central concern, consider the descriptive comparisons reported in Table 3, based on NHIS 2009 data and reproduced from Angrist and Pischke (2015, Table 1.1).

	Husbands			Wives		
	Some HI	No HI	Δ	Some HI	No HI	Δ
Health	4.01	3.70	0.31 (0.03)	4.02	3.62	0.39 (0.04)
Nonwhite	0.16	0.17	-0.01 (0.01)	0.15	0.17	-0.02 (0.01)
Age	43.98	41.26	2.71 (0.23)	42.24	39.62	2.62 (0.30)
Educ.	14.31	11.56	2.74 (0.10)	14.44	11.80	2.64 (0.11)
Family size	3.50	3.98	-0.47 (0.05)	3.49	3.93	-0.43 (0.05)
Employed	0.92	0.85	0.07 (0.01)	0.77	0.56	0.21 (0.02)

Table 3: Descriptive differences by health insurance status

These comparisons show substantial differences between individuals with some health insurance and those with no health insurance. For both husbands and wives, insured individuals report better health, are older, have more education, and are more likely to be employed.

This is exactly the kind of pattern that generates concern about selection bias. Insurance status is clearly related to many characteristics that are also related to outcomes. Therefore, a raw comparison between insured and uninsured individuals does not isolate the causal effect of insurance.

Remark. The important point is not merely that the two groups differ in observables. Rather, differences in observables suggest that they may also differ in unobservables and, more fundamentally, in their potential outcomes. That is precisely what undermines a causal interpretation of simple mean comparisons.

Intuition from the insurance example

Suppose we compare the health of insured and uninsured individuals and find that the insured are healthier on average. It would be incorrect to immediately conclude that insurance improves health by that amount. People who obtain insurance may already differ systematically from those who do not: they may have different jobs, income, education, family structure, or health risks.

So the observed difference mixes together two things:

1. the causal effect of insurance, and
2. the pre-existing differences between those who end up insured and those who do not.

This is the essence of selection bias.

A final intuition on marginal distributions

Under random assignment, we identify the marginal distributions

$$F_{Y(1)}(y) \quad \text{and} \quad F_{Y(0)}(y).$$

The word “marginal” means that we study each potential outcome distribution separately.

The treated group reveals the distribution of outcomes under treatment because, when $D = 1$, we observe

$$Y = Y(1).$$

Likewise, the control group reveals the distribution of outcomes under no treatment because, when $D = 0$, we observe

$$Y = Y(0).$$

Therefore, the experiment lets us recover the two distributions corresponding to the two policy regimes:

- the distribution of outcomes if everyone were treated,
- the distribution of outcomes if no one were treated.

From these two marginal distributions, we can compute many causal objects such as

$$ATE, \quad DTE, \quad QTE.$$

However, even in an experiment, we still do not observe both $Y(1)$ and $Y(0)$ for the same unit. Therefore, we do not identify the joint distribution of

$$(Y(0), Y(1)),$$

and hence we do not automatically identify the full distribution of individual treatment effects

$$Y(1) - Y(0).$$

Remark. This is a very useful distinction. Random assignment tells us what the treated world looks like and what the untreated world looks like, each taken separately. But it does not tell us how the two worlds are paired within the same individual.

An analogy for marginal distributions

Think of two histograms. One histogram describes outcomes in the world where everyone receives treatment; the other describes outcomes in the world where no one receives treatment. Random assignment allows us to estimate both histograms.

But to know the treatment effect for each individual, we would need to know which treated

outcome and untreated outcome belong to the same person. That matching is unobserved. So experiments identify each distribution separately, but not the full individual-level mapping between them.

Unconfoundedness and the Causal Interpretation of Regression

A common informal justification for giving a causal interpretation to linear regression is the claim that we “control for all relevant factors.” The goal of this section is to formalize that argument within the potential outcomes framework.

The key idea is to begin with a causal estimand and an identification assumption, and only then ask when regression can recover that estimand. The central assumption introduced here is the *unconfoundedness* assumption.

Remark. The phrase “controlling for all relevant factors” is not, by itself, a formal identification argument. In the potential outcomes framework, this intuition is translated into a conditional independence restriction: after conditioning on observed covariates, treatment assignment is independent of the relevant potential outcomes.

Roadmap

The discussion proceeds in two steps:

1. First, we study identification of treatment effects under unconfoundedness.
2. Second, we revisit linear regression through the lens of the potential outcomes framework under the same assumption.

The emphasis is therefore not primarily on regression as a statistical routine, but on the causal conditions under which regression admits a causal interpretation.

References

Main references:

- Imbens (2004)
- Angrist and Pischke (2009)
- Wooldridge (2010)
- Imbens and Rubin (2015)

Additional reference:

- Angrist and Pischke (2015)

Identification under Unconfoundedness: Core Assumptions

Identification of treatment effects under selection on observables rests on three ingredients:

1. **SUTVA**,
2. **Unconfoundedness**,
3. **Overlap**.

These assumptions play different roles. SUTVA ensures that potential outcomes are well-defined. Unconfoundedness delivers conditional exogeneity of treatment assignment. Overlap guarantees that treated and untreated units are both observed for the relevant values of the covariates.

Remark. It is useful to separate these assumptions conceptually. Unconfoundedness alone is not enough for identification if, for some values of the covariates, treatment status is deterministic. In that case there is no valid within- X comparison.

Unconfoundedness

Let $D \in \{0, 1\}$ denote treatment status, let X be a vector of observed covariates, and let $Y(1)$ and $Y(0)$ denote the potential outcomes under treatment and control. The unconfoundedness assumption is

$$(Y(1), Y(0)) \perp D \mid X.$$

This assumption is also known as:

- ignorable treatment assignment,
- selection on observables,
- conditional independence.

Interpretation. Conditional on $X = x$, treatment assignment is as good as random. Thus, among units with the same observable characteristics, treated and untreated units are comparable in terms of their potential outcomes.

Equivalently, for each $d \in \{0, 1\}$,

$$Y(d) \perp D \mid X.$$

Hence,

$$\mathbb{E}[Y(d) \mid D = 1, X = x] = \mathbb{E}[Y(d) \mid D = 0, X = x].$$

Intuition. The assumption does *not* require treatment to be randomly assigned unconditionally. There may be substantial selection into treatment. What it requires is that all variables jointly affecting treatment choice and potential outcomes are contained in X . After conditioning on X , no residual selection bias remains.

Remark. A useful informal summary is:

“Selection may exist, but only through observables.”

That is the precise content of selection on observables.

Unconfoundedness and Exogeneity in the Linear Regression Model

A useful question is how unconfoundedness relates to the exogeneity assumption in the classical linear regression model.

Start from the potential-outcomes identity

$$Y = DY(1) + (1 - D)Y(0).$$

Suppose that treatment effects are constant:

$$Y(1) - Y(0) = \tau$$

for some constant τ . Then

$$Y(1) = Y(0) + \tau,$$

so that

$$Y = Y(0) + \tau D.$$

Now assume that the untreated potential outcome is linear in observables:

$$Y(0) = X'\beta + e.$$

Substituting into the previous expression yields

$$Y = \tau D + X'\beta + e.$$

Thus, under constant treatment effects and an additive linear specification for $Y(0)$, the observed outcome satisfies a linear regression equation.

Key equivalence. Under these assumptions,

$$(Y(0), Y(1)) \perp D \mid X \quad \Longleftrightarrow \quad e \perp D \mid X.$$

Why? Since $Y(1) = Y(0) + \tau$, the pair $(Y(0), Y(1))$ is fully determined by $Y(0)$. Therefore,

$$(Y(0), Y(1)) \perp D \mid X \iff Y(0) \perp D \mid X.$$

But

$$Y(0) = X'\beta + e,$$

and conditional on X , the term $X'\beta$ is non-random. Hence the only remaining source of variation in $Y(0)$ is e , so

$$Y(0) \perp D \mid X \iff e \perp D \mid X.$$

Remark. This result shows that, in this special linear-additive constant-effect setting, unconfoundedness is equivalent to conditional exogeneity of the regression error. Outside this setting, the two concepts need not coincide.

Example: Technology Adoption

Consider a firm deciding whether to adopt a new technology $D \in \{0, 1\}$. Output is given by

$$Y = g(D, e),$$

so that the potential outcomes are

$$Y(d) = g(d, e), \quad d \in \{0, 1\}.$$

Let C denote the cost of adoption. If the firm adopts decision d , profits are

$$\pi(d) = g(d, e) - Cd.$$

The firm chooses the action that maximizes expected profits conditional on C :

$$D = \arg \max_{d \in \{0, 1\}} \mathbb{E}[g(d, e) - Cd \mid C].$$

If

$$C \perp e,$$

then the adoption cost is independent of the unobserved productivity component. In that case,

$$\mathbb{E}[g(d, e) - Cd \mid C] = \mathbb{E}[g(d, e)] - Cd,$$

and therefore the adoption rule becomes

$$D = \mathbf{1}\left\{\mathbb{E}[g(1, e) - g(0, e)] \geq C\right\} = h(C).$$

Implication for output. The output potential outcomes are

$$Y(0) = g(0, e), \quad Y(1) = g(1, e),$$

which depend only on e , whereas treatment choice D depends only on C . Since $C \perp e$, it follows that

$$(Y(0), Y(1)) \perp D.$$

Hence treatment is unconfounded for the outcome Y .

Implication for profits. The profit potential outcomes are

$$\pi(d) = g(d, e) - Cd,$$

which depend on both e and C . Since treatment choice D is itself a function of C , the potential profits are generally not independent of treatment status. Therefore unconfoundedness fails for profits.

This shows that unconfoundedness is outcome-specific: the same treatment assignment may be unconfounded for output, but not for profits.

Remark. The core intuition is that firms select into treatment based on adoption cost C . This does not create bias for output if C is unrelated to the output shocks e . But it does create bias for profits, because C enters profits directly.

Overlap and Regression-Based Identification

Define the propensity score by

$$p(X) \equiv \mathbb{E}[D \mid X] = \Pr(D = 1 \mid X).$$

Since $D \in \{0, 1\}$, the conditional expectation of D coincides with the conditional treatment probability.

Overlap. The overlap assumption is

$$0 < p(X) < 1.$$

This is also called *common support* or *positivity*.

Interpretation. For every relevant value of X , there must be both treated and untreated units. Thus treatment assignment must be non-degenerate within covariate cells.

Remark. Unconfoundedness says that treated and untreated units are causally comparable conditional on X . Overlap says that such comparisons are empirically feasible in the data.

Why overlap matters. If $p(x) = 1$, then all units with $X = x$ are treated, so $\mathbb{E}[Y(0) \mid X = x]$ cannot be learned from the data. Similarly, if $p(x) = 0$, then $\mathbb{E}[Y(1) \mid X = x]$ is not identified. Thus overlap rules out support failures that would make the relevant counterfactual mean unobservable.

Regression-Based Identification

Suppose that unconfoundedness and overlap hold. Define the observable regression functions

$$m_1(x) \equiv \mathbb{E}[Y \mid X = x, D = 1], \quad m_0(x) \equiv \mathbb{E}[Y \mid X = x, D = 0].$$

Then:

$$ATE = \mathbb{E}[m_1(X) - m_0(X)],$$

and

$$ATT = \mathbb{E}[m_1(X) - m_0(X) \mid D = 1].$$

Step 1: identification of the conditional effect. By consistency,

$$Y = Y(1) \quad \text{when } D = 1, \quad Y = Y(0) \quad \text{when } D = 0.$$

Therefore,

$$\mathbb{E}[Y \mid X = x, D = 1] = \mathbb{E}[Y(1) \mid X = x, D = 1],$$

and

$$\mathbb{E}[Y \mid X = x, D = 0] = \mathbb{E}[Y(0) \mid X = x, D = 0].$$

Under unconfoundedness,

$$\mathbb{E}[Y(1) \mid X = x, D = 1] = \mathbb{E}[Y(1) \mid X = x],$$

and

$$\mathbb{E}[Y(0) \mid X = x, D = 0] = \mathbb{E}[Y(0) \mid X = x].$$

Hence,

$$m_1(x) - m_0(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

So the difference in conditional observed means identifies the conditional average treatment effect:

$$CATE(x) \equiv \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

Step 2: integration over the relevant distribution of X . Applying the law of iterated expectations,

$$ATE = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[\mathbb{E}[Y(1) - Y(0) \mid X]] = \mathbb{E}[m_1(X) - m_0(X)].$$

Similarly,

$$ATT = \mathbb{E}[Y(1) - Y(0) \mid D = 1] = \mathbb{E}[\mathbb{E}[Y(1) - Y(0) \mid X] \mid D = 1] = \mathbb{E}[m_1(X) - m_0(X) \mid D = 1].$$

Remark. The difference between the two formulas is in the weighting distribution. The ATE averages over the unconditional distribution of X , while the ATT averages over the distribution of X among treated units.

A Weaker Assumption for Average Effects

If we are only interested in average effects, full unconfoundedness can be weakened to conditional mean independence:

$$\mathbb{E}[Y(d) \mid D, X] = \mathbb{E}[Y(d) \mid X], \quad d \in \{0, 1\}.$$

This is weaker than

$$(Y(1), Y(0)) \perp D \mid X,$$

because it only requires equality of conditional means, not equality of the full conditional distributions.

Remark. Full unconfoundedness is stronger than needed for identifying average treatment effects. For mean effects, it is enough that treatment assignment be conditionally exogenous in mean.

The Propensity Score Theorem

A key result due to Rosenbaum and Rubin is that under unconfoundedness it is sufficient to condition on the scalar propensity score $p(X)$, rather than on the entire vector X .

Proposition (Propensity score theorem). *Suppose unconfoundedness holds:*

$$(Y(1), Y(0)) \perp D \mid X.$$

Then

$$(Y(1), Y(0)) \perp D \mid p(X).$$

Proof. Because $D \in \{0, 1\}$, it is enough to show that

$$P(D = 1 \mid Y(1), Y(0), p(X)) = P(D = 1 \mid p(X)).$$

Start with the left-hand side. Since D is binary,

$$P(D = 1 \mid Y(1), Y(0), p(X)) = \mathbb{E}[D \mid Y(1), Y(0), p(X)].$$

Using iterated expectations,

$$\mathbb{E}[D \mid Y(1), Y(0), p(X)] = \mathbb{E}[\mathbb{E}[D \mid Y(1), Y(0), p(X), X] \mid Y(1), Y(0), p(X)].$$

Because $p(X)$ is a measurable function of X , conditioning on $(p(X), X)$ is the same as conditioning on X , so

$$\mathbb{E}[D \mid Y(1), Y(0), p(X), X] = \mathbb{E}[D \mid Y(1), Y(0), X].$$

Now apply unconfoundedness:

$$(Y(1), Y(0)) \perp D \mid X \implies \mathbb{E}[D \mid Y(1), Y(0), X] = \mathbb{E}[D \mid X].$$

By definition of the propensity score,

$$\mathbb{E}[D \mid X] = P(D = 1 \mid X) = p(X).$$

Therefore,

$$P(D = 1 \mid Y(1), Y(0), p(X)) = \mathbb{E}[p(X) \mid Y(1), Y(0), p(X)] = p(X).$$

On the other hand,

$$P(D = 1 \mid p(X)) = \mathbb{E}[D \mid p(X)] = \mathbb{E}[\mathbb{E}[D \mid X] \mid p(X)] = \mathbb{E}[p(X) \mid p(X)] = p(X).$$

Hence

$$P(D = 1 \mid Y(1), Y(0), p(X)) = P(D = 1 \mid p(X)),$$

which proves

$$(Y(1), Y(0)) \perp D \mid p(X).$$

□

Remark. This theorem is a dimensionality-reduction result. If X is high-dimensional, directly conditioning on X may be difficult. The propensity score theorem shows that, for identification purposes, the relevant information in X can be summarized by the scalar $p(X)$.

Remark. The theorem does *not* say that any one-dimensional function of X is enough. It says

that the specific object

$$p(X) = P(D = 1 \mid X)$$

contains all the information in X that matters for treatment assignment.

Identification Based on Inverse Probability Weighting (IPW)

The propensity score leads directly to a useful class of identification formulas and estimators. The idea of IPW is simple: units that are unlikely to receive the treatment status they actually received are given larger weight, because they represent a larger fraction of the relevant population.

Proposition (IPW identification). *Suppose unconfoundedness and overlap hold. Then*

$$ATE = \mathbb{E} \left[\frac{(D - p(X))Y}{p(X)(1 - p(X))} \right] = \mathbb{E} \left[\frac{DY}{p(X)} \right] - \mathbb{E} \left[\frac{(1 - D)Y}{1 - p(X)} \right].$$

Moreover,

$$ATT = \mathbb{E} \left[\frac{(D - p(X))Y}{P(D = 1)(1 - p(X))} \right].$$

Proof. We prove the two formulas separately.

Step 1: ATE. Using the observed-outcome identity

$$Y = DY(1) + (1 - D)Y(0),$$

we have

$$DY = DY(1), \quad (1 - D)Y = (1 - D)Y(0).$$

Consider the treated component:

$$\mathbb{E} \left[\frac{DY}{p(X)} \right] = \mathbb{E} \left[\frac{DY(1)}{p(X)} \right].$$

Apply iterated expectations conditioning on X :

$$\mathbb{E} \left[\frac{DY(1)}{p(X)} \right] = \mathbb{E} \left[\frac{1}{p(X)} \mathbb{E}[DY(1) \mid X] \right].$$

Under unconfoundedness,

$$Y(1) \perp D \mid X,$$

so

$$\begin{aligned}
\mathbb{E}[DY(1) \mid X] &= \mathbb{E}[D \mid X]\mathbb{E}[Y(1) \mid X] \\
&= \mathbb{E}[DY(1) \mid X, D = 1] \cdot P(D = 1 \mid X) + \underbrace{\mathbb{E}[DY(1) \mid X, D = 0]}_{=0} \cdot P(D = 0 \mid X) \\
&= p(X)\mathbb{E}[Y(1) \mid X].
\end{aligned}$$

Therefore,

$$\mathbb{E}\left[\frac{DY}{p(X)}\right] = \mathbb{E}[\mathbb{E}[Y(1) \mid X]] = \mathbb{E}[Y(1)].$$

Similarly,

$$\mathbb{E}\left[\frac{(1-D)Y}{1-p(X)}\right] = \mathbb{E}\left[\frac{(1-D)Y(0)}{1-p(X)}\right].$$

Again conditioning on X ,

$$\mathbb{E}\left[\frac{(1-D)Y(0)}{1-p(X)}\right] = \mathbb{E}\left[\frac{1}{1-p(X)}\mathbb{E}[(1-D)Y(0) \mid X]\right].$$

Because $Y(0) \perp D \mid X$,

$$\mathbb{E}[(1-D)Y(0) \mid X] = (1-p(X))\mathbb{E}[Y(0) \mid X].$$

Hence

$$\mathbb{E}\left[\frac{(1-D)Y}{1-p(X)}\right] = \mathbb{E}[Y(0)].$$

Subtracting the two identities yields

$$\mathbb{E}\left[\frac{DY}{p(X)}\right] - \mathbb{E}\left[\frac{(1-D)Y}{1-p(X)}\right] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = ATE.$$

Finally, since

$$\frac{DY}{p(X)} - \frac{(1-D)Y}{1-p(X)} = \frac{(D-p(X))Y}{p(X)(1-p(X))},$$

we obtain

$$ATE = \mathbb{E}\left[\frac{(D-p(X))Y}{p(X)(1-p(X))}\right].$$

Step 2: ATT. Recall that

$$ATT = \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

We identify the two terms separately.

First,

$$\mathbb{E}[Y(1) \mid D = 1] = \frac{\mathbb{E}[DY(1)]}{P(D = 1)} = \frac{\mathbb{E}[DY]}{P(D = 1)}.$$

Second, consider

$$\mathbb{E} \left[\frac{p(X)(1-D)Y}{P(D=1)(1-p(X))} \right] = \frac{1}{P(D=1)} \mathbb{E} \left[\frac{p(X)(1-D)Y(0)}{1-p(X)} \right].$$

Conditioning on X ,

$$\frac{1}{P(D=1)} \mathbb{E} \left[\frac{p(X)}{1-p(X)} \mathbb{E}[(1-D)Y(0) | X] \right].$$

Using $Y(0) \perp D | X$,

$$\mathbb{E}[(1-D)Y(0) | X] = (1-p(X))\mathbb{E}[Y(0) | X],$$

so the expression becomes

$$\frac{1}{P(D=1)} \mathbb{E}[p(X)\mathbb{E}[Y(0) | X]].$$

Since $\mathbb{E}[D | X] = p(X)$ and $Y(0) \perp D | X$,

$$p(X)\mathbb{E}[Y(0) | X] = \mathbb{E}[D | X]\mathbb{E}[Y(0) | X] = \mathbb{E}[DY(0) | X].$$

Therefore,

$$\frac{1}{P(D=1)} \mathbb{E}[p(X)\mathbb{E}[Y(0) | X]] = \frac{\mathbb{E}[DY(0)]}{P(D=1)} = \mathbb{E}[Y(0) | D=1].$$

Hence

$$ATT = \frac{\mathbb{E}[DY]}{P(D=1)} - \mathbb{E} \left[\frac{p(X)(1-D)Y}{P(D=1)(1-p(X))} \right].$$

Combining the two terms over the common denominator $P(D=1)(1-p(X))$, we obtain

$$ATT = \mathbb{E} \left[\frac{(D-p(X))Y}{P(D=1)(1-p(X))} \right].$$

□

Remark. The ATE formula reweights treated units by $1/p(X)$ and untreated units by $1/(1-p(X))$. Units that receive a treatment status that was unlikely given their covariates get more weight, because they are especially informative about the counterfactual mean.

Remark. For the ATT, the weighting is asymmetric. Since the target parameter is the average treatment effect *for the treated*, the treated group enters directly, while the control group is reweighted so as to reproduce the covariate distribution of the treated population.

Remark. Overlap is crucial for IPW. If $p(X)$ is very close to 0 or 1, the weights

$$\frac{1}{p(X)} \quad \text{and} \quad \frac{1}{1-p(X)}$$

become very large. This foreshadows an important practical issue: even when identification

holds, IPW estimators can be unstable when overlap is weak.

Interpreting Regression under Unconfoundedness

We now connect the unconfoundedness framework to linear regression. There are two useful interpretations.

Interpretation 1: Regression as the best linear approximation

Under unconfoundedness,

$$(Y(1), Y(0)) \perp D \mid X,$$

the observed conditional mean has a causal interpretation:

$$\mathbb{E}[Y \mid D = d, X] = \mathbb{E}[Y(d) \mid X], \quad d \in \{0, 1\}.$$

Equivalently,

$$\mathbb{E}[Y \mid D, X] = D \mathbb{E}[Y(1) \mid X] + (1 - D) \mathbb{E}[Y(0) \mid X].$$

Thus, when we run a linear regression of Y on D and X , we may interpret it as constructing the *best linear approximation* to a causal conditional expectation function.

Formally, the population linear regression coefficient vector solves

$$(\tau, \beta) = \arg \min_{t, b} \mathbb{E}[(Y - tD - X'b)^2].$$

Hence the linear predictor

$$\tau D + X'\beta$$

is the best linear predictor of $\mathbb{E}[Y \mid D, X]$, and under unconfoundedness this means it is the best linear approximation to the causal object

$$D \mathbb{E}[Y(1) \mid X] + (1 - D) \mathbb{E}[Y(0) \mid X].$$

Remark. This interpretation is useful but limited. Regression does not generally recover the exact causal response surface unless that surface is itself linear in (D, X) . What regression delivers is the best linear summary of that surface.

Remark. A common mistake is to conclude that the coefficient on D must equal the ATE. That is not true in general. Without additional structure, linear regression typically recovers a particular weighted average of conditional treatment effects, not the simple unweighted average.

Interpretation 2: Regression as a weighted average of treatment effects

Consider the population linear regression model

$$Y = \tau D + X' \beta + e,$$

where (τ, β) are defined by the population least-squares projection of Y on (D, X) .

The following result characterizes the coefficient τ under unconfoundedness.

Proposition (Interpreting linear regression). *Suppose unconfoundedness holds:*

$$(Y(1), Y(0)) \perp D \mid X.$$

Suppose further that

$$\mathbb{E}[D \mid X] = X' \pi.$$

Then

$$\tau = \frac{\mathbb{E}[\sigma_D^2(X) ATE(X)]}{\mathbb{E}[\sigma_D^2(X)]},$$

where

$$ATE(X) \equiv \mathbb{E}[Y(1) - Y(0) \mid X]$$

and

$$\sigma_D^2(X) \equiv \mathbb{E}[(D - \mathbb{E}[D \mid X])^2 \mid X] = \mathbf{Var}(D \mid X).$$

Proof. Let

$$\tilde{D} \equiv D - \mathbb{E}[D \mid X].$$

Because $\mathbb{E}[D \mid X] = X' \pi$ is linear in X , the Frisch–Waugh–Lovell theorem implies that the population coefficient on D in the regression of Y on (D, X) can be written as

$$\tau = \frac{\mathbb{E}[\tilde{D} Y]}{\mathbb{E}[\tilde{D}^2]}.$$

We now analyze the numerator. Using

$$Y = Y(0) + D(Y(1) - Y(0)),$$

we obtain

$$\mathbb{E}[\tilde{D} Y] = \mathbb{E}[\tilde{D} Y(0)] + \mathbb{E}[\tilde{D} D(Y(1) - Y(0))].$$

Consider the first term. By iterated expectations,

$$\mathbb{E}[\tilde{D} Y(0)] = \mathbb{E}[\mathbb{E}[\tilde{D} Y(0) \mid X]].$$

Under unconfoundedness, $Y(0) \perp D \mid X$, so conditional on X ,

$$\mathbb{E}[\tilde{D} Y(0) \mid X] = \mathbb{E}[\tilde{D} \mid X] \mathbb{E}[Y(0) \mid X] = 0,$$

since $\mathbb{E}[\tilde{D} \mid X] = 0$. Therefore,

$$\mathbb{E}[\tilde{D} Y(0)] = 0.$$

Now consider the second term. Again by iterated expectations,

$$\mathbb{E}[\tilde{D} D(Y(1) - Y(0))] = \mathbb{E}[\mathbb{E}[\tilde{D} D(Y(1) - Y(0)) \mid X]].$$

Under unconfoundedness,

$$Y(1) - Y(0) \perp D \mid X,$$

so

$$\mathbb{E}[\tilde{D} D(Y(1) - Y(0)) \mid X] = \mathbb{E}[\tilde{D} D \mid X] \mathbb{E}[Y(1) - Y(0) \mid X].$$

That is,

$$\mathbb{E}[\tilde{D} D(Y(1) - Y(0)) \mid X] = \mathbb{E}[\tilde{D} D \mid X] ATE(X).$$

Since $\tilde{D} = D - \mathbb{E}[D \mid X]$,

$$\mathbb{E}[\tilde{D} D \mid X] = \mathbb{E}[D^2 \mid X] - \mathbb{E}[D \mid X]^2.$$

Because $D \in \{0, 1\}$, we have $D^2 = D$, so

$$\mathbb{E}[\tilde{D} D \mid X] = \mathbb{E}[D \mid X] - \mathbb{E}[D \mid X]^2 = \mathbf{Var}(D \mid X) = \sigma_D^2(X).$$

Hence

$$\mathbb{E}[\tilde{D} Y] = \mathbb{E}[\sigma_D^2(X) ATE(X)].$$

For the denominator,

$$\mathbb{E}[\tilde{D}^2] = \mathbb{E}[\mathbb{E}[\tilde{D}^2 \mid X]] = \mathbb{E}[\mathbf{Var}(D \mid X)] = \mathbb{E}[\sigma_D^2(X)].$$

Combining numerator and denominator,

$$\tau = \frac{\mathbb{E}[\sigma_D^2(X) ATE(X)]}{\mathbb{E}[\sigma_D^2(X)]}.$$

This proves the result. □

Remark. This proposition shows that regression estimates a *variance-weighted average* of condi-

tional treatment effects. The weights are

$$w(X) = \frac{\sigma_D^2(X)}{\mathbb{E}[\sigma_D^2(X)]}, \quad w(X) \geq 0, \quad \mathbb{E}[w(X)] = 1.$$

So τ is an average of $ATE(X)$, but not generally the simple average.

Remark. When treatment is binary,

$$\sigma_D^2(X) = \mathbf{Var}(D \mid X) = p(X)(1 - p(X)).$$

Therefore regression puts more weight on values of X where treatment assignment is more variable—that is, where treated and untreated units coexist in substantial numbers. By contrast, cells with $p(X)$ close to 0 or 1 receive little weight.

Remark. This weighting formula is very intuitive. Identification under unconfoundedness comes from comparing treated and untreated units with the same X . If for some covariate profile almost everyone is treated or almost no one is treated, there is little within- X variation in treatment status, so linear regression learns relatively little from that region.

Remark. If treatment effects are constant, so that

$$ATE(X) = \tau_0 \quad \text{for all } X,$$

then the weighting becomes irrelevant and the regression coefficient satisfies

$$\tau = \tau_0.$$

Thus, heterogeneity in treatment effects is exactly what makes the regression coefficient differ from the ATE.

Remark. The assumption

$$\mathbb{E}[D \mid X] = X' \pi$$

holds automatically if the regression is saturated in X . In that case, the linear projection of D on X coincides with the true conditional expectation.

Direct Estimation as an Alternative to Linear Regression

An alternative to linear regression is to estimate causal parameters such as the ATE and ATT directly.

The identification results under unconfoundedness are constructive: once we write the target parameters as functions of observable conditional expectations and probabilities, we can replace those population objects by sample analogues. This leads to several important classes of estimators.

- **Regression adjustment:** estimate $\mathbb{E}[Y \mid D = d, X]$ and then average the implied counterfactual outcomes.
- **Inverse probability weighting (IPW):** estimate the propensity score $p(X)$ and use reweighting formulas such as those derived above.
- **Doubly robust methods:** combine outcome regression and propensity score weighting.

Remark. Doubly robust estimators are especially attractive because they combine two models: one for the outcome and one for treatment assignment. In their most basic form, they remain consistent if at least one of these two models is correctly specified.

Remark. This perspective is conceptually important. Linear regression is not the only route from unconfoundedness to estimation. It is one possible approximation strategy. IPW and doubly robust estimators instead target the causal parameter more directly.

Remark. This direct-estimation perspective also opens the door to modern machine learning methods. One may estimate nuisance objects such as $\mathbb{E}[Y \mid D, X]$ and $p(X)$ flexibly, and then plug them into orthogonal or doubly robust estimators. This is the logic behind the modern treatment of causal inference in high-dimensional settings.

4.1 A comment on the inclusion of covariates

Consider the regression model

$$Y = \tau D + X'\beta + e,$$

where

$$(X, Y(1), Y(0)) \perp D$$

i.e. a completely randomized experiment. Then,

$$\tau = ATE = ATT \quad \text{Theorem 7.X in Imbens and Rubin (2015).}$$

So, if D and X are independent, there is not omitted variable bias, so including X only matter for efficiency.

But here we have

$$(Y(1), Y(0)) \perp D \mid X.$$

that is conditioned on X (unconfoundedness), then,

$$\tau = \mathbb{E}[\omega(X)ATE(X)] \text{ for } \omega(X) \geq 0$$

provided that $\mathbb{E}[D|X]$ is linear.

The Local Average Treatment Effects (LATE) Framework

References. Main references for this topic include Vytlačil (2002), Angrist and Pischke (2009), Huber and Wüthrich (2019), and Blandhol et al. (2025). Additional references include Abadie (2003), Angrist and Pischke (2015), Hansen (2022), and the handbook treatment by Mogstad and Torgovitsky (2024).

Introduction

A central question in instrumental variables analysis is the following:

What does 2SLS/IV identify when treatment effects are heterogeneous?

Suppose D and Z are binary, and consider the simple linear IV model

$$Y = \alpha + \beta D + e.$$

If Z is used as an instrument for D , then the IV estimand can be written as

$$\beta^{IV} = \frac{\text{Cov}(Y, Z)}{\text{Cov}(D, Z)} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]}.$$

Thus, the key issue is how to interpret this ratio when the causal effect of D on Y is not constant across individuals.

The answer is provided by the LATE framework of Imbens and Angrist (1994): under appropriate assumptions, the IV/Wald estimand identifies the average treatment effect for a specific subpopulation, namely the *compliers*.

Potential Outcomes and Potential Treatments

Consider a setting with a general outcome Y , a binary treatment D , and a binary instrument Z . For now, we abstract from covariates X .

A leading special case is a randomized experiment with imperfect compliance. The instrument Z may be an assignment indicator, while D is actual treatment take-up.

To describe this environment, it is useful to define both potential outcomes and potential treatments.

Potential outcomes. Index the potential outcomes by $(d, z) \in \{0, 1\}^2$. Thus, for each unit there are four potential outcomes:

$$Y(1, 1), \quad Y(1, 0), \quad Y(0, 1), \quad Y(0, 0).$$

Potential treatments. Treatment status may itself depend on the instrument. Hence we define two potential treatments:

$$D(1), \quad D(0),$$

where $D(z)$ denotes the treatment the unit would receive if the instrument were set equal to z .

Observed variables. The realized treatment and realized outcome satisfy

$$D = ZD(1) + (1 - Z)D(0),$$

and

$$Y = ZY(D(1), 1) + (1 - Z)Y(D(0), 0).$$

Remark. This framework is richer than the standard treatment-effects setup because the outcome may depend both on the treatment received and directly on the instrument. The exclusion restriction introduced below rules out such direct effects of Z on Y .

Four Types

The pair $(D(0), D(1))$ determines how a unit reacts to the instrument. This yields four behavioral types:

$$\begin{array}{ll} \text{Always-takers} & (T = a) : \quad D(0) = 1, D(1) = 1, \\ \text{Never-takers} & (T = n) : \quad D(0) = 0, D(1) = 0, \\ \text{Compliers} & (T = c) : \quad D(0) = 0, D(1) = 1, \\ \text{Defiers} & (T = df) : \quad D(0) = 1, D(1) = 0. \end{array}$$

Remark. Always-takers and never-takers do not respond to the instrument. Compliers move their treatment status in the same direction as the instrument. Defiers move in the opposite direction.

Remark. The LATE parameter will eventually be interpreted as the average treatment effect for compliers. Thus, the type classification is not just descriptive; it is fundamental for understanding what IV identifies.

LATE Assumptions

We now state the standard assumptions of the LATE framework.

Definition (LATE assumptions). The LATE framework imposes the following conditions:

1. Independence:

$$(Y(1, 1), Y(1, 0), Y(0, 1), Y(0, 0), D(1), D(0)) \perp Z.$$

2. **Exclusion:**

$$P(Y(d,0) = Y(d,1)) = 1, \quad d \in \{0,1\}.$$

3. **First stage:**

$$P(Z = 1) \in (0,1), \quad \mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0] \neq 0.$$

4. **Monotonicity:**

$$P(D(1) \geq D(0)) = 1.$$

Discussion.

Remark. Independence does *not* imply exclusion. Independence says that the instrument is as-good-as-random with respect to all potential outcomes and potential treatments. Exclusion says something different: conditional on the treatment level d , changing the instrument from 0 to 1 does not directly affect the outcome.

Remark. Under exclusion, it is legitimate to suppress the dependence of the outcome on z and write

$$Y(d) = Y(d, z), \quad z \in \{0,1\}.$$

In other words, once treatment status is fixed at d , the instrument no longer matters for the outcome.

Remark. Monotonicity is the key new assumption in the LATE framework. It rules out defiers. Equivalently, it requires that the instrument weakly increases treatment take-up for every unit.

Remark. Under independence,

$$\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0] = \mathbb{E}[D(1) - D(0)].$$

To see this, note that

$$\mathbb{E}[D \mid Z = z] = \mathbb{E}[D(z)]$$

by independence, so

$$\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0] = \mathbb{E}[D(1)] - \mathbb{E}[D(0)] = \mathbb{E}[D(1) - D(0)].$$

Remark. Under monotonicity, $D(1) - D(0) \in \{0,1\}$, and

$$D(1) - D(0) = 1 \iff D(1) > D(0),$$

which occurs exactly for compliers. Hence

$$\mathbb{E}[D(1) - D(0)] = P(D(1) > D(0)).$$

Therefore, under independence and monotonicity,

$$\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0] = P(D(1) > D(0)).$$

So the first stage is exactly the proportion of compliers in the population.

Remark. This last identity provides a very useful intuition: if the instrument barely changes treatment take-up, then there are very few compliers, and the IV design has little identifying content for treatment effects.

Examples of Potential Failures of LATE Assumptions

Example (Violation of exclusion). Consider the draft lottery application studied by Angrist (1990). Let

$$Z_i = 1\{\text{individual } i \text{ was draft eligible}\}, \quad D_i = 1\{\text{individual } i \text{ is a veteran}\},$$

and let Y_i denote earnings.

The intended instrumental-variables logic is that draft eligibility affects earnings only through veteran status. Under the exclusion restriction, this means that for each treatment status $d \in \{0, 1\}$,

$$Y_i(d, 1) = Y_i(d, 0) \quad \text{a.s.}$$

That is, once we hold fixed whether the individual becomes a veteran, draft eligibility should have no additional direct effect on earnings.

A potential concern is that this exclusion restriction may fail because of *educational draft deferments*. If draft-eligible individuals had an incentive to remain in college longer in order to avoid induction, then draft eligibility could affect schooling decisions. Since schooling itself may affect later earnings, the instrument could influence the outcome through a channel other than veteran status:

$$Z_i \longrightarrow \text{schooling} \longrightarrow Y_i.$$

If this channel is present, then the reduced-form effect of Z_i on Y_i would reflect not only the causal effect of veteran status, but also the effect of changes in education induced by draft eligibility. In that case, the IV/Wald estimand would no longer isolate the causal effect of D_i on Y_i .

The economic point is simple: exogeneity of the instrument is not enough. Even if draft eligibility is as-good-as-random, the exclusion restriction can still fail if draft eligibility changes earnings through any mechanism other than veteran status.

Empirically, Angrist and Krueger (1992) found little evidence of a substantial relationship between draft lottery eligibility and schooling, in part because educational deferments were being phased out during the relevant period. Thus, this concern is conceptually important

even if it may not be quantitatively important in that specific application.

Remark. This example is useful because it clarifies the difference between *independence* and *exclusion*. Independence says that the instrument is exogenous. Exclusion says that the instrument operates only through the treatment of interest. These are logically distinct requirements.

Example (Violation of monotonicity). Consider the setting in Duflo and Saez (2003). Let

$$Z_i = 1\{\text{individual } i \text{ receives a financial incentive}\},$$

$$D_i = 1\{\text{individual } i \text{ attends the information meeting}\},$$

and let Y_i denote take-up of the retirement plan.

The monotonicity assumption requires that the instrument weakly increases treatment take-up for every individual:

$$D_i(1) \geq D_i(0) \quad \text{a.s.}$$

Equivalently, there are no defiers, that is, no individuals for whom

$$D_i(1) = 0, \quad D_i(0) = 1.$$

Why might monotonicity fail here? A possible concern is that financial incentives may *crowd out intrinsic motivation*. Some individuals may have attended the information meeting in the absence of payment because they were already interested in retirement planning. Once a financial incentive is introduced, however, the meeting may be perceived differently: it may look less informative, more promotional, or simply less appealing. If this crowding-out effect is strong enough, some individuals who would have attended without the incentive may choose not to attend with the incentive.

Such individuals are precisely defiers:

$$D_i(0) = 1, \quad D_i(1) = 0.$$

If a non-negligible fraction of the population behaves this way, then monotonicity fails.

The intuition is that monotonicity is not merely a statistical regularity about average attendance. It is a *unit-level* requirement: the instrument must move each person weakly in the same direction. Even if the incentive raises attendance on average, monotonicity can still fail if it discourages attendance for some individuals.

This type of concern is discussed more generally by de Chaisemartin (2017), who emphasizes that monotonicity can be substantively demanding in applications where incentives may generate heterogeneous behavioral responses.

Remark. The key distinction is between an instrument that increases treatment take-up for *most* individuals and an instrument that increases take-up for *every* individual. LATE requires

the latter in weak form, namely that the instrument does not induce anyone to move in the opposite direction.

Vytlacil's Equivalence Result

Vytlacil (2002) shows that the LATE assumptions admit a simple structural interpretation. In particular, when the instrument and treatment are binary, the LATE framework is equivalent to a threshold-crossing model for treatment selection.

Proposition (Vytlacil's equivalence result). *Suppose $Z \in \{0, 1\}$ and $D \in \{0, 1\}$, and impose the LATE assumptions: independence, exclusion, a nonzero first stage, and monotonicity. Then there exist a latent variable U and a non-trivial function $v(\cdot)$ such that*

$$D = 1\{v(Z) \geq U\}, \quad Z \perp (U, Y(1), Y(0)).$$

Conversely, any model of this form satisfies the LATE assumptions.

For a binary instrument, non-triviality simply means that

$$v(1) \neq v(0).$$

Without loss of generality, we may assume that

$$v(1) > v(0).$$

Remark. The latent variable U can be interpreted as an individual's resistance, cost, or reluctance to take the treatment. The function $v(Z)$ measures how strongly the instrument pushes the individual toward treatment. Thus, the treatment rule

$$D = 1\{v(Z) \geq U\}$$

says that an individual takes the treatment whenever the push generated by the instrument is large enough to overcome his or her latent resistance.

Under the threshold-crossing representation, the potential treatment statuses are

$$D(0) = 1\{v(0) \geq U\}, \quad D(1) = 1\{v(1) \geq U\}.$$

Since $v(1) > v(0)$, the possible compliance types are determined entirely by the location of U :

- **Always-takers:**

$$U \leq v(0).$$

In this case,

$$D(0) = 1, \quad D(1) = 1.$$

- **Compliers:**

$$v(0) < U \leq v(1).$$

In this case,

$$D(0) = 0, \quad D(1) = 1.$$

- **Never-takers:**

$$v(1) < U.$$

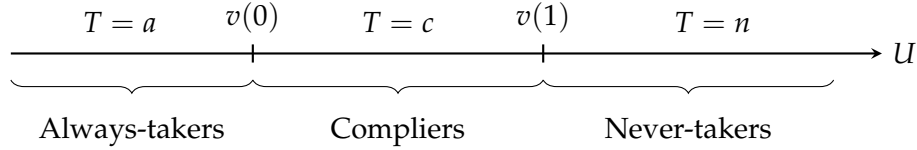
In this case,

$$D(0) = 0, \quad D(1) = 0.$$

Remark. This representation makes monotonicity transparent. Since $v(1) > v(0)$, we automatically have

$$D(1) \geq D(0) \quad \text{a.s.}$$

Hence defiers cannot exist in a threshold-crossing model of this form. The compliers are exactly those individuals whose latent resistance U lies between the two thresholds $v(0)$ and $v(1)$.



Remark. The figure provides the key intuition behind Vytlačil's result. The instrument shifts the treatment threshold from $v(0)$ to $v(1)$. Individuals far to the left have such low resistance that they always take treatment. Individuals far to the right have such high resistance that they never take treatment. Only those in the intermediate region switch treatment status when the instrument changes, and these are precisely the compliers.

The LATE Theorem

We now state the main identification result of the LATE framework.

Proposition (LATE Theorem). *Suppose the LATE assumptions hold. Then*

$$\frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]} = \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0)].$$

That is, the Wald estimand identifies the average treatment effect for compliers:

$$LATE := \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0)].$$

Proof. Under exclusion, the potential outcome depends only on treatment status, so we may

write

$$Y(1) = Y(1,1) = Y(1,0), \quad Y(0) = Y(0,1) = Y(0,0).$$

Hence, using consistency,

$$Y = Y(0) + D(Y(1) - Y(0)), \quad D = ZD(1) + (1 - Z)D(0).$$

Fix $z \in \{0,1\}$. Then

$$\mathbb{E}[Y \mid Z = z] = \mathbb{E}[Y(0) + D(z)(Y(1) - Y(0)) \mid Z = z].$$

By independence of

$$(Y(1), Y(0), D(1), D(0)) \perp Z,$$

we can drop the conditioning on $Z = z$, so

$$\mathbb{E}[Y \mid Z = z] = \mathbb{E}[Y(0) + D(z)(Y(1) - Y(0))].$$

Therefore,

$$\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] = \mathbb{E}[(D(1) - D(0))(Y(1) - Y(0))].$$

Next, again by independence,

$$\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0] = \mathbb{E}[D(1) - D(0)].$$

Now use monotonicity:

$$D(1) \geq D(0) \quad \text{a.s.}$$

Since $D(1), D(0) \in \{0,1\}$, this implies

$$D(1) - D(0) = 1\{D(1) > D(0)\}.$$

Thus,

$$\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] = \mathbb{E}[(Y(1) - Y(0)) 1\{D(1) > D(0)\}],$$

and

$$\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0] = P(D(1) > D(0)).$$

Because the first stage is nonzero, we have

$$P(D(1) > D(0)) > 0.$$

Dividing numerator and denominator yields

$$\begin{aligned}\frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]} &= \frac{\mathbb{E}[(Y(1) - Y(0)) 1\{D(1) > D(0)\}]}{P(D(1) > D(0))} \\ &= \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0)].\end{aligned}$$

This proves the result. □

Remark. The theorem shows that IV does *not* identify the average treatment effect in the whole population in general. Rather, it identifies the average treatment effect for the subpopulation of *compliers*, namely those individuals whose treatment status is changed by the instrument.

Remark. The size of the complier subpopulation is identified by the first stage:

$$P(D(1) > D(0)) = \mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0].$$

Hence, if the first stage is small, the complier group may be a very small fraction of the population.

Remark. This result gives the fundamental interpretation of the Wald estimator in the presence of heterogeneous treatment effects: it is an average causal effect, but only for an instrument-specific margin of individuals.

Remark. Is the LATE a useful quantity? There is disagreement on this. See, for example, the papers in the issue in *JEL* Vol. 48, No. 2.³

The LATE vs. Other Average Effects

A natural question is how the LATE relates to other average treatment effects. In general, the LATE is neither equal to the ATT nor to the ATE.

Recall that under monotonicity the population can be partitioned into three types:

Always-takers: $D(0) = D(1) = 1$,

Compliers: $D(0) = 0, D(1) = 1$,

Never-takers: $D(0) = D(1) = 0$.

Thus,

$$\text{LATE} = \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0)] = \mathbb{E}[Y(1) - Y(0) \mid T = c].$$

LATE $\neq$ ATT in general. The average treatment effect on the treated is

$$\text{ATT} := \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

³<https://www.aeaweb.org/issues/140>

But the treated population is not, in general, the same as the complier population. Indeed, under monotonicity,

$$D = 1 \implies \text{either } T = a, \text{ or } (T = c \text{ and } Z = 1).$$

Therefore,

$$\begin{aligned} \mathbb{E}[Y(1) - Y(0) \mid D = 1] &= \mathbb{E}[Y(1) - Y(0) \mid T = c] P(T = c, Z = 1 \mid D = 1) \\ &\quad + \mathbb{E}[Y(1) - Y(0) \mid T = a] P(T = a \mid D = 1). \end{aligned}$$

Hence the ATT is a weighted average of the treatment effects for compliers and always-takers among the treated. In general, this is different from the LATE, which averages only over compliers.

LATE $\neq$ ATE in general. The average treatment effect in the full population is

$$\text{ATE} := \mathbb{E}[Y(1) - Y(0)].$$

Using the partition of the population into always-takers, compliers, and never-takers, we obtain

$$\begin{aligned} \mathbb{E}[Y(1) - Y(0)] &= \mathbb{E}[Y(1) - Y(0) \mid T = c] P(T = c) \\ &\quad + \mathbb{E}[Y(1) - Y(0) \mid T = a] P(T = a) \\ &\quad + \mathbb{E}[Y(1) - Y(0) \mid T = n] P(T = n). \end{aligned}$$

Thus the ATE averages treatment effects over the entire population, whereas the LATE averages only over compliers. Unless these group-specific average treatment effects coincide, the ATE and the LATE will differ.

Remark. The distinction can also be seen directly from the observed (D, Z) cells:

$$D = 1, Z = 0 \implies T = a,$$

$$D = 1, Z = 1 \implies T \in \{a, c\},$$

$$D = 0, Z = 1 \implies T = n,$$

$$D = 0, Z = 0 \implies T \in \{n, c\}.$$

So the treated group is not the same as the complier group, and the full population is larger still.

Remark. Therefore, the LATE is an instrument-specific average treatment effect: it captures the

causal effect for those individuals whose treatment status is changed by the instrument, not necessarily for the treated population and not necessarily for the full population.

Remark. See Abadie (2003) for weighting representations that relate the LATE to other treatment effect parameters.

Covariates

We now extend the LATE framework to allow for observed covariates X .

Definition (LATE- X assumptions). The LATE framework with covariates imposes the following conditions:

1. **Independence:**

$$(Y(1,1), Y(1,0), Y(0,1), Y(0,0), D(1), D(0)) \perp\!\!\!\perp Z \mid X.$$

2. **Exclusion:**

$$P(Y(d,0) = Y(d,1) \mid X) = 1, \quad d \in \{0,1\}.$$

3. **First stage:**

$$P(Z = 1 \mid X) \in (0,1), \quad \mathbb{E}[D \mid Z = 1, X] - \mathbb{E}[D \mid Z = 0, X] \neq 0.$$

4. **Monotonicity:**

$$P(D(1) \geq D(0) \mid X) = 1.$$

Remark. These are the direct analogues of the usual LATE assumptions, but now imposed conditional on the observed covariates X . Thus, the identifying argument goes through within covariate cells.

Proposition (Identification of the conditional LATE). *Under the LATE- X assumptions, the conditional local average treatment effect is identified for every covariate value x such that*

$$\mathbb{E}[D \mid Z = 1, X = x] - \mathbb{E}[D \mid Z = 0, X = x] \neq 0.$$

Specifically,

$$\text{LATE}(x) := \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0), X = x]$$

is given by

$$\text{LATE}(x) = \frac{\mathbb{E}[Y \mid Z = 1, X = x] - \mathbb{E}[Y \mid Z = 0, X = x]}{\mathbb{E}[D \mid Z = 1, X = x] - \mathbb{E}[D \mid Z = 0, X = x]}.$$

Remark. The proof is exactly the same as for the unconditional LATE theorem, except that every step is carried out conditional on $X = x$.

Identification of the Unconditional LATE

The unconditional LATE can be written as the average of the conditional LATEs over the distribution of X among compliers:

$$\text{LATE} = \int \text{LATE}(x) dF_{X|D(1) > D(0)}(x).$$

Remark. This expression shows that the unconditional LATE is not the simple average of the conditional LATEs with respect to the marginal distribution of X . Rather, it averages $\text{LATE}(x)$ using the covariate distribution of the complier population.

Proposition (Frölich's representation of the unconditional LATE). *Suppose the LATE-X assumptions hold. Then the unconditional LATE can be written as*

$$\text{LATE} = \frac{\int (\mathbb{E}[Y | Z = 1, X = x] - \mathbb{E}[Y | Z = 0, X = x]) dF_X(x)}{\int (\mathbb{E}[D | Z = 1, X = x] - \mathbb{E}[D | Z = 0, X = x]) dF_X(x)}.$$

Proof. Let

$$C := 1\{D(1) > D(0)\}$$

denote the complier indicator. Then

$$\text{LATE} = \mathbb{E}[Y(1) - Y(0) | C = 1] = \int \text{LATE}(x) dF_{X|C=1}(x).$$

By Bayes' rule,

$$dF_{X|C=1}(x) = \frac{P(C = 1 | X = x)}{P(C = 1)} dF_X(x).$$

Hence

$$\text{LATE} = \frac{1}{P(C = 1)} \int \text{LATE}(x) P(C = 1 | X = x) dF_X(x).$$

Now, under conditional independence and conditional monotonicity,

$$P(C = 1 | X = x) = \mathbb{E}[D | Z = 1, X = x] - \mathbb{E}[D | Z = 0, X = x].$$

Also, by the conditional LATE theorem,

$$\text{LATE}(x) = \frac{\mathbb{E}[Y | Z = 1, X = x] - \mathbb{E}[Y | Z = 0, X = x]}{\mathbb{E}[D | Z = 1, X = x] - \mathbb{E}[D | Z = 0, X = x]}.$$

Therefore,

$$\text{LATE}(x) P(C = 1 | X = x) = \mathbb{E}[Y | Z = 1, X = x] - \mathbb{E}[Y | Z = 0, X = x].$$

Substituting into the previous display gives

$$\text{LATE} = \frac{\int (\mathbb{E}[Y \mid Z = 1, X = x] - \mathbb{E}[Y \mid Z = 0, X = x]) dF_X(x)}{P(C = 1)}.$$

Finally,

$$P(C = 1) = \int P(C = 1 \mid X = x) dF_X(x) = \int (\mathbb{E}[D \mid Z = 1, X = x] - \mathbb{E}[D \mid Z = 0, X = x]) dF_X(x),$$

which yields the stated formula. $\square$

Remark. This representation is useful because it shows that the unconditional LATE is directly identified from conditional reduced-form and first-stage objects. Hence it can be estimated without imposing a linear IV model.

Remark. See Frölich (2007) for this identification result, and Huber and Wüthrich (2019) for references on estimation and inference.

IV/2SLS = LATE?

IV/2SLS with Covariates

The LATE theorem shows that IV/2SLS identifies the LATE in the binary-instrument, binary-treatment setup *without* covariates. We now study what happens when covariates are included linearly in the IV specification.

More specifically, consider an IV regression of Y on D and X , using Z as an instrument for D . Assume

$$D \in \{0, 1\}, \quad Z \in \{0, 1\},$$

and that X includes a constant.

Using the Frisch–Waugh–Lovell theorem, Blandhol et al. (2025) write the IV estimand as

$$\beta_{iv} = \frac{\mathbb{E}(Y\tilde{Z})}{\mathbb{E}(D\tilde{Z})}, \quad \tilde{Z} = Z - \mathcal{P}(Z \mid X),$$

where $\mathcal{P}(Z \mid X)$ denotes the linear projection of Z on X .

To state the result, define the shorthand

$$\Delta_c(x) := \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0), X = x],$$

$$\Delta_a(x) := \mathbb{E}[Y(1) - Y(0) \mid D(1) = D(0) = 1, X = x].$$

Proposition (Proposition 1 in Blandhol et al. (2025)). *Suppose that*

$$\mathbb{E}(Y(d) \mid X) = X' \eta_d, \quad d \in \{0, 1\}.$$

Then

$$\beta_{iv} = \mathbb{E}[\omega_c(X)\Delta_c(X)] + \mathbb{E}[\omega_a(X)\Delta_a(X)],$$

where the weights are

$$\omega_c(X) := \mathbb{E}(Z | X)(1 - \mathcal{P}(Z | X))P(D(1) > D(0) | X) \mathbb{E}(\tilde{Z}D)^{-1},$$

and

$$\omega_a(X) := \mathbb{E}(\tilde{Z} | X) P(D(1) = D(0) = 1 | X) \mathbb{E}(\tilde{Z}D)^{-1}.$$

Proof. Let

$$m(X) := \mathbb{E}(Z | X), \quad p(X) := \mathcal{P}(Z | X), \quad \mu(X) := \mathbb{E}(\tilde{Z} | X) = m(X) - p(X),$$

and define

$$P_c(X) := P(D(1) > D(0) | X), \quad P_a(X) := P(D(1) = D(0) = 1 | X).$$

Recall that

$$\beta_{iv} = \frac{\mathbb{E}(Y\tilde{Z})}{\mathbb{E}(D\tilde{Z})}.$$

Thus it suffices to derive a decomposition for the numerator $\mathbb{E}(Y\tilde{Z})$.

Step 1: Decompose the numerator. Using iterated expectations,

$$\mathbb{E}(Y\tilde{Z}) = \mathbb{E}[\mathbb{E}(Y\tilde{Z} | X)].$$

Now use the identity

$$\mathbb{E}(AB | X) = \mathbf{Cov}(A, B | X) + \mathbb{E}(A | X)\mathbb{E}(B | X),$$

with $A = Y$ and $B = \tilde{Z}$, to obtain

$$\mathbb{E}(Y\tilde{Z}) = \underbrace{\mathbb{E}[\mathbf{Cov}(Y, \tilde{Z} | X)]}_{=:A} + \underbrace{\mathbb{E}[\mathbb{E}(Y | X)\mathbb{E}(\tilde{Z} | X)]}_{=:B}.$$

Since $\tilde{Z} = Z - \mathcal{P}(Z | X)$ and $\mathcal{P}(Z | X)$ is measurable with respect to X ,

$$\mathbf{Cov}(Y, \tilde{Z} | X) = \mathbf{Cov}(Y, Z | X).$$

Therefore,

$$A = \mathbb{E}[\mathbf{Cov}(Y, Z | X)].$$

Step 2: Evaluate term A.

We first recall a useful identity. For any random variable W and any binary variable $B \in$

$\{0, 1\}$,

$$\mathbf{Cov}(W, B \mid X) = (\mathbb{E}(W \mid B = 1, X) - \mathbb{E}(W \mid B = 0, X))\mathbb{E}(B \mid X)(1 - \mathbb{E}(B \mid X)).$$

Applying this with $W = Y$ and $B = Z$, we get

$$\mathbf{Cov}(Y, Z \mid X) = (\mathbb{E}(Y \mid Z = 1, X) - \mathbb{E}(Y \mid Z = 0, X))m(X)(1 - m(X)).$$

Now, conditional on X , the LATE theorem yields

$$\mathbb{E}(Y \mid Z = 1, X) - \mathbb{E}(Y \mid Z = 0, X) = \Delta_c(X) \left(\mathbb{E}(D \mid Z = 1, X) - \mathbb{E}(D \mid Z = 0, X) \right).$$

Under monotonicity,

$$\mathbb{E}(D \mid Z = 1, X) - \mathbb{E}(D \mid Z = 0, X) = P_c(X).$$

Hence

$$\mathbf{Cov}(Y, Z \mid X) = \Delta_c(X)P_c(X)m(X)(1 - m(X)).$$

Therefore,

$$A = \mathbb{E}[\Delta_c(X)P_c(X)m(X)(1 - m(X))].$$

Step 3: Evaluate term B .

By monotonicity, conditional on X , the population is partitioned into always-takers, compliers, and never-takers. Hence

$$\begin{aligned} \mathbb{E}(Y \mid X) &= \mathbb{E}(Y \mid T = a, X)P(T = a \mid X) \\ &\quad + \mathbb{E}(Y \mid T = n, X)P(T = n \mid X) \\ &\quad + \mathbb{E}(Y \mid T = c, X)P(T = c \mid X). \end{aligned}$$

Using the type-specific treatment behavior,

$$Y = Y(1) \quad \text{for always-takers,} \quad Y = Y(0) \quad \text{for never-takers,}$$

and

$$Y = ZY(1) + (1 - Z)Y(0) \quad \text{for compliers.}$$

Therefore,

$$\begin{aligned} \mathbb{E}(Y \mid X) &= \mathbb{E}(Y(1) \mid T = a, X)P_a(X) \\ &\quad + \mathbb{E}(Y(0) \mid T = n, X)P(T = n \mid X) \\ &\quad + \mathbb{E}[ZY(1) + (1 - Z)Y(0) \mid T = c, X] P_c(X). \end{aligned}$$

By conditional independence of Z ,

$$Z \perp\!\!\!\perp (Y(1), Y(0), D(1), D(0)) \mid X,$$

so

$$\mathbb{E}[ZY(1) + (1 - Z)Y(0) \mid T = c, X] = m(X)\mathbb{E}(Y(1) \mid T = c, X) + (1 - m(X))\mathbb{E}(Y(0) \mid T = c, X).$$

Thus

$$\begin{aligned} \mathbb{E}(Y \mid X) &= \mathbb{E}(Y(1) \mid T = a, X)P_a(X) \\ &\quad + \mathbb{E}(Y(0) \mid T = n, X)P(T = n \mid X) \\ &\quad + \mathbb{E}(Y(0) \mid T = c, X)P_c(X) + m(X)\Delta_c(X)P_c(X). \end{aligned}$$

Now add and subtract $\mathbb{E}(Y(0) \mid T = a, X)P_a(X)$. This gives

$$\mathbb{E}(Y \mid X) = \mathbb{E}(Y(0) \mid X) + \Delta_a(X)P_a(X) + m(X)\Delta_c(X)P_c(X).$$

By assumption,

$$\mathbb{E}(Y(0) \mid X) = X'\eta_0.$$

Hence

$$\mathbb{E}(Y \mid X) = X'\eta_0 + \Delta_a(X)P_a(X) + m(X)\Delta_c(X)P_c(X).$$

Substituting this into B , we obtain

$$\begin{aligned} B &= \mathbb{E}[(X'\eta_0 + \Delta_a(X)P_a(X) + m(X)\Delta_c(X)P_c(X))\mu(X)] \\ &= \mathbb{E}[\Delta_a(X)P_a(X)\mu(X)] + \mathbb{E}[m(X)\Delta_c(X)P_c(X)\mu(X)] \\ &\quad + \mathbb{E}[X'\eta_0 \mu(X)]. \end{aligned}$$

Finally,

$$\mathbb{E}[X'\eta_0 \mu(X)] = \eta_0'\mathbb{E}[X\tilde{Z}] = 0,$$

because $\tilde{Z} = Z - \mathcal{P}(Z \mid X)$ is the residual from the linear projection of Z on X , and is therefore orthogonal to X . Thus

$$B = \mathbb{E}[\Delta_a(X)P_a(X)\mu(X)] + \mathbb{E}[m(X)\Delta_c(X)P_c(X)\mu(X)].$$

Step 4: Sum the two terms.

Combining A and B ,

$$\begin{aligned}\mathbb{E}(Y\tilde{Z}) &= \mathbb{E}\left[\Delta_c(X)P_c(X)\left(m(X)(1-m(X)) + m(X)\mu(X)\right)\right] \\ &\quad + \mathbb{E}[\Delta_a(X)P_a(X)\mu(X)].\end{aligned}$$

Since $\mu(X) = m(X) - p(X)$,

$$m(X)(1-m(X)) + m(X)\mu(X) = m(X)(1-p(X)).$$

Therefore,

$$\mathbb{E}(Y\tilde{Z}) = \mathbb{E}[\Delta_c(X)P_c(X)m(X)(1-p(X))] + \mathbb{E}[\Delta_a(X)P_a(X)\mu(X)].$$

Dividing by $\mathbb{E}(\tilde{Z}D)$, we obtain

$$\beta_{iv} = \mathbb{E}[\omega_c(X)\Delta_c(X)] + \mathbb{E}[\omega_a(X)\Delta_a(X)],$$

where

$$\omega_c(X) = m(X)(1-p(X))P_c(X)\mathbb{E}(\tilde{Z}D)^{-1},$$

and

$$\omega_a(X) = \mu(X)P_a(X)\mathbb{E}(\tilde{Z}D)^{-1}.$$

Recalling that

$$m(X) = \mathbb{E}(Z | X), \quad p(X) = \mathcal{P}(Z | X), \quad \mu(X) = \mathbb{E}(\tilde{Z} | X),$$

this is exactly the claimed result. □

Remark (Why the sign conclusions about the weights follow). The conclusions about the weights follow from two simple observations.

1. Since $Z \in \{0, 1\}$,

$$\mathbb{E}(Z | X) \in [0, 1].$$

2. Also,

$$\mathbb{E}[\mathbb{E}(\tilde{Z} | X)] = \mathbb{E}(\tilde{Z}) = 0.$$

Therefore, either

$$P(\mathbb{E}(\tilde{Z} | X) = 0) = 1,$$

or $\mathbb{E}(\tilde{Z} | X)$ must take both positive and negative values.

Hence, if $\mathbb{E}(\tilde{Z}D) > 0$, the sign of $\omega_c(X)$ is governed by the factor

$$1 - \mathcal{P}(Z | X),$$

while $\omega_a(X)$ inherits the sign changes of $\mathbb{E}(\tilde{Z} | X)$. In particular, unless $\mathbb{E}(\tilde{Z} | X) = 0$ almost surely, the always-taker component gives rise to negative weights for some values of X .

Remark. This result shows that, once covariates are included linearly, the IV estimand need not equal the LATE. Instead, it is generally a weighted average of

1. complier treatment effects, through $\Delta_c(X)$, and
2. always-taker treatment effects, through $\Delta_a(X)$.

Thus, linear IV with covariates need not recover a purely complier-specific causal parameter.

Remark. If $\mathbb{E}(\tilde{Z}D) > 0$, then

$$\omega_c(X) < 0 \quad \text{if and only if} \quad \mathcal{P}(Z | X) > 1.$$

Moreover,

$$\omega_a(X) < 0$$

with positive probability unless

$$\mathbb{E}(\tilde{Z} | X) = 0.$$

Remark. Because

$$\mathbb{E}(\tilde{Z} | X) = \mathbb{E}(Z | X) - \mathcal{P}(Z | X),$$

whenever

$$\mathbb{E}(Z | X) \neq \mathcal{P}(Z | X),$$

there are negatively weighted effects for some groups.

Implications for Applied Research

The result has several implications for practice.

1. If possible, use flexible specifications that ensure

$$\mathbb{E}(Z | X) = \mathcal{P}(Z | X).$$

2. If sufficiently flexible specifications are not possible, test whether

$$\mathbb{E}(Z | X) = \mathcal{P}(Z | X),$$

for example using Ramsey's RESET test.

3. If $Z \in \{0, 1\}$, check that

$$\mathcal{P}(Z \mid X) \in [0, 1].$$

This is a necessary, though not sufficient, condition for

$$\mathbb{E}(Z \mid X) = \mathcal{P}(Z \mid X).$$

4. Use alternative estimators; see Section 5 in Blandhol et al. (2025).

Remark. The key lesson is that IV/2SLS with linearly included covariates is not automatically a LATE estimator. The linear specification for covariate adjustment matters, and misspecification can generate contamination by always-taker effects as well as negative weighting.

5 Introduction to Panel Data Analysis

Panel data consist of repeated observations on the same units over multiple periods of time.⁴ This type of data structure is ubiquitous in modern empirical economics.

It is often useful to distinguish between two broad types of panel data:

1. **Micro panels:** panels with a large number of units and a relatively small number of time periods.
2. **Macro panels:** panels with a moderate number of units and often a moderate to large number of time periods.

In micro panels, and often also in macro panels, a common working assumption is that observations are independent across units, while observations within a given unit may be serially dependent over time. That is, the data are allowed to be autocorrelated within unit, but dependence across different units is ruled out.

This assumption induces a natural *cluster structure*: each unit forms a cluster, and observations within the same cluster may be correlated. As a consequence, standard inference procedures that treat all observations as mutually independent are generally inappropriate. This is the main reason why panel-data applications often rely on *cluster-robust standard errors*, typically clustering at the unit level.

5.1 Why panel data matter

The key feature of panel data is that they allow us to track the same unit over time. Relative to a purely cross-sectional dataset, this gives us additional leverage to study dynamic behavior, persistent heterogeneity across units, and the role of within-unit variation over time.

⁴Main references for this section are Jeffrey M. Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. (MIT Press, 2010), and Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022).

At the same time, the repeated-observation structure creates econometric complications. In particular, because observations within a unit are typically dependent, valid inference must account for within-unit correlation.

5.2 Balanced and Unbalanced Panels

Units are indexed by $i = 1, \dots, N$, and time periods are indexed by $t = 1, \dots, T$.

We distinguish between two standard panel-data structures:

1. **Balanced panels:** panels in which data are available for all units in all time periods.
2. **Unbalanced panels:** panels in which some observations are missing, so that not every unit is observed in every period.

In an unbalanced panel, unit i is observed for $T_i \leq T$ periods. Let $S_i \subseteq \{1, \dots, T\}$ denote the set of time periods for which unit i is observed.

The total number of observations depends on whether the panel is balanced or unbalanced. In a balanced panel, we have

$$n = NT.$$

In an unbalanced panel, the total sample size is

$$n = \sum_{i=1}^N T_i.$$

5.3 Notation

We index observations by the pair (i, t) . For each unit i and time period t , let

$$Y_{it} \in \mathbb{R}$$

denote the outcome variable, and let

$$X_{it} \in \mathbb{R}^k$$

denote the vector of covariates.

To simplify the exposition, it is convenient to introduce stacked vector and matrix notation at the unit level. For each unit i , define:

$$\begin{aligned} Y_i &:= \text{the } T_i \times 1 \text{ vector stacking } Y_{it}, \quad t \in S_i, \text{ in chronological order,} \\ X_i &:= \text{the } T_i \times k \text{ matrix stacking } X'_{it}, \quad t \in S_i, \text{ in chronological order.} \end{aligned}$$

Thus,

$$Y_i = \begin{pmatrix} Y_{it_1} \\ Y_{it_2} \\ \vdots \\ Y_{it_{T_i}} \end{pmatrix}, \quad X_i = \begin{pmatrix} X'_{it_1} \\ X'_{it_2} \\ \vdots \\ X'_{it_{T_i}} \end{pmatrix}, \quad \text{where } S_i = \{t_1 < t_2 < \cdots < t_{T_i}\}.$$

Finally, stacking across units, define the full-sample objects

$$Y = (Y'_1, \dots, Y'_N)' \quad \text{and} \quad X = (X'_1, \dots, X'_N)'.$$

This notation is especially useful in panel-data analysis because it makes explicit the two-level structure of the data: observations are stacked over time within units, and then across units in the sample.

5.4 Pooled Regression Model

The pooled regression model essentially ignores the panel structure. In particular, it treats the panel as a single sample of observations and abstracts from the fact that repeated observations belong to the same unit.

Using the notation introduced above, the model can be written at three different levels:

$$\begin{aligned} Y_{it} &= X'_{it}\beta + e_{it}, \\ Y_i &= X_i\beta + e_i, \\ Y &= X\beta + e. \end{aligned}$$

The corresponding OLS estimator, called the *pooled regression estimator*, is

$$\begin{aligned} \hat{\beta}_{\text{pool}} &= \left(\sum_{i=1}^N \sum_{t \in S_i} X_{it} X'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t \in S_i} X_{it} Y_{it} \right) \\ &= \left(\sum_{i=1}^N X'_i X_i \right)^{-1} \left(\sum_{i=1}^N X'_i Y_i \right) \\ &= (X'X)^{-1}(X'Y). \end{aligned}$$

Thus, the pooled estimator is numerically identical to the usual OLS estimator obtained by stacking all observations and running a single regression. What makes panel data special is therefore not the algebra of the estimator itself, but rather the dependence structure of the data and the assumptions under which this estimator is meaningful.

5.5 Strict Mean Independence

A standard justification for $\hat{\beta}_{\text{pool}}$ is the assumption of *strict mean independence*:

$$E(e_{it} \mid X_i) = 0,$$

where X_i denotes the full collection of covariates for unit i across all periods.

This assumption requires that the error in period t , e_{it} , be mean independent of all covariates for unit i , including past, present, and future values. Hence, strict mean independence is substantially stronger than a contemporaneous exogeneity condition.

An important implication is that lagged outcomes cannot be included among the regressors. To see this, suppose

$$X_{it} = Y_{i,t-1}.$$

Then, at date $t + 1$, the regressor becomes

$$X_{i,t+1} = Y_{it}.$$

But from the regression equation,

$$Y_{it} = X'_{it}\beta + e_{it},$$

so $X_{i,t+1}$ necessarily contains e_{it} . Therefore, $X_{i,t+1}$ is correlated with e_{it} , which violates strict mean independence.

Interpretation. The pooled model is attractive because it is simple and uses standard OLS machinery. However, its validity rests on a strong exogeneity requirement. In panel settings, this requirement is often too restrictive, which is one reason why fixed-effects and related estimators become important.

5.6 One-Way Error Component Model

A natural extension of the pooled regression model is the *one-way error component model*:

$$Y_{it} = X'_{it}\beta + e_{it} = X'_{it}\beta + u_i + \varepsilon_{it}.$$

In this specification, the composite error e_{it} has two components:

1. u_i : a time-invariant unit-specific component or effect;
2. ε_{it} : an idiosyncratic shock.

Thus, the term u_i captures persistent heterogeneity across units, while ε_{it} captures transitory variation within a unit over time.

At the unit level, the model can be written as

$$Y_i = X_i\beta + \mathbf{1}_i u_i + \varepsilon_i,$$

where

$$\mathbf{1}_i = (1, \dots, 1)' \in \mathbb{R}^{T_i}.$$

This representation makes clear that the unit-specific effect u_i enters each observation for unit i in exactly the same way, because it is constant over time.

5.7 Random Effects vs. Fixed Effects

There are two leading approaches to the one-way error component model. They differ in the assumptions imposed on the relationship between the unit-specific effect u_i and the regressors X_{it} .

1. **Random effects model:** u_i and X_{it} are uncorrelated.
2. **Fixed effects model:** u_i may be arbitrarily correlated with X_{it} .

The distinction is econometrically important. If u_i is correlated with the regressors, then the pooled regression estimator is inconsistent. Under the same circumstance, the random effects estimator is also inconsistent.⁵

Because correlation between u_i and X_{it} is often plausible in applications, fixed effects estimators are frequently preferred.

Interpretation. The key issue is whether the unobserved time-invariant heterogeneity u_i can be treated as exogenous. If it can, then random effects is a viable strategy. If it cannot, then the panel structure must be exploited more aggressively to eliminate u_i , which is exactly the logic behind fixed effects methods.

5.8 Two-Way Error Component Model

An important generalization of the one-way error component model is the *two-way error component model*:

$$Y_{it} = X_{it}'\beta + v_t + u_i + \varepsilon_{it}.$$

This specification augments the one-way error component model by introducing a time effect v_t . Thus, the model allows for two distinct sources of unobserved heterogeneity:

1. a unit-specific effect u_i , which is constant over time for a given unit;
2. a time-specific effect v_t , which is common across units in period t .

The term u_i captures time-invariant heterogeneity across units, while v_t captures aggregate shocks or common factors that vary over time but affect all units in a given period. The remaining component ε_{it} is the idiosyncratic shock.

The econometric appeal of this formulation is that it allows us to control simultaneously for unobserved heterogeneity across units and for common time shocks. In particular, it is

⁵See Section 17.6 in Hansen (2022).

possible to estimate this model without imposing assumptions on the relationship between X_{it} and v_t by using *two-way fixed effects* estimators.

6

5.9 Fixed Effects Model: Assumptions and Interpretation

We now focus on the fixed effects (FE) version of the one-way error component model. The model is

$$Y_{it} = X'_{it}\beta + u_i + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T.$$

The key assumption underlying FE estimation is *strict exogeneity* of the idiosyncratic error:

$$E(X_{is}\varepsilon_{it}) = 0 \quad \text{for all } s = 1, \dots, T.$$

Equivalently, in a stronger conditional form, one often writes

$$E(\varepsilon_{it} \mid X_i, u_i) = 0,$$

where X_i denotes the full history of regressors for unit i .

The condition

$$E(X_{is}\varepsilon_{it}) = 0 \quad \text{for all } s = 1, \dots, T$$

is called *strict exogeneity*. It requires that the idiosyncratic shock in period t be orthogonal to all regressors for the same unit, including past, present, and future values.

Why is this less restrictive than pooled OLS? The fixed effects model is less restrictive than pooled OLS in one specific and important dimension: it allows the time-invariant unobserved effect u_i to be arbitrarily correlated with the regressors.

Under pooled OLS, the composite error is

$$e_{it} = u_i + \varepsilon_{it},$$

and the relevant exogeneity requirement is essentially

$$E(e_{it} \mid X_i) = 0,$$

or equivalently,

$$E(u_i + \varepsilon_{it} \mid X_i) = 0.$$

Thus pooled OLS requires both u_i and ε_{it} to be orthogonal to the regressors.

By contrast, FE estimation is designed precisely so that u_i can be correlated with X_i . The identifying assumption is imposed only on the idiosyncratic component ε_{it} , not on the time-

⁶For further discussion, see Chapter 17.27 in Hansen (2022).

invariant heterogeneity u_i .

Hence, FE relaxes the exogeneity requirement with respect to unobserved time-invariant heterogeneity. This is the key value-added of panel data methods relative to pooled cross-sectional regression.

Remark. The FE model is less restrictive than pooled OLS because it permits

$$\text{Cov}(u_i, X_{it}) \neq 0.$$

However, it does *not* relax the temporal dimension of exogeneity: strict exogeneity still requires that ε_{it} be orthogonal to all regressors X_{is} , past, present, and future. For this reason, standard FE estimation typically rules out lagged dependent variables as regressors unless further methods are introduced.⁷

5.10 Within Transformation

The main insight underlying FE estimation is that the unit-specific effect u_i can be removed by a suitable transformation. We start with the *within transformation*.

For any variable A_{it} , define its unit-specific time average by

$$\bar{A}_i = \frac{1}{T_i} \sum_{t \in S_i} A_{it},$$

and define the demeaned variable

$$\ddot{A}_{it} = A_{it} - \bar{A}_i.$$

Applying this to the FE model

$$Y_{it} = X'_{it}\beta + u_i + \varepsilon_{it},$$

and averaging over time for a fixed unit i , we obtain

$$\bar{Y}_i = \bar{X}'_i\beta + u_i + \bar{\varepsilon}_i.$$

Subtracting this time average from the original equation yields

$$Y_{it} - \bar{Y}_i = (X_{it} - \bar{X}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

Equivalently,

$$\ddot{Y}_{it} = \ddot{X}'_{it}\beta + \ddot{\varepsilon}_{it}.$$

This shows that the within transformation removes u_i , because u_i is constant over time for

⁷See, for example, Bonhomme, S. (2025), *Back to Feedback: Dynamics and Heterogeneity in Panel Data*.

a given unit:

$$u_i - u_i = 0.$$

Therefore, any omitted variable bias originating from time-invariant unobserved heterogeneity is eliminated by this transformation. This is the fundamental logic behind FE estimation.

Why does this work? The argument relies on three crucial ingredients:

1. **Linearity:** the model must be linear in the additive effect u_i ;
2. **Time-invariance of β :** the slope coefficient must be common across periods;
3. **Time-invariance of u_i :** the unobserved effect must not vary over time.

If any of these conditions fail, the within transformation will generally not isolate β in this simple way.

Remark. The within transformation does not eliminate all sources of endogeneity. It removes only the time-invariant component u_i . If the idiosyncratic shocks ε_{it} are correlated with the regressors, then FE estimation remains inconsistent. Thus FE solves omitted variable bias coming from fixed unobserved heterogeneity, but not endogeneity arising from time-varying omitted factors, simultaneity, or feedback from past shocks into future regressors.

5.11 Matrix Notation for the Within Transformation

It is useful to express the within transformation in matrix form. For any unit-level vector A_i , define its demeaned version by

$$\ddot{A}_i = A_i - \mathbf{1}_i \bar{A}_i,$$

where

$$\bar{A}_i = (\mathbf{1}'_i \mathbf{1}_i)^{-1} \mathbf{1}'_i A_i.$$

Since $\mathbf{1}'_i \mathbf{1}_i = T_i$, this is simply the average of the entries of A_i . Hence,

$$\ddot{A}_i = A_i - \mathbf{1}_i (\mathbf{1}'_i \mathbf{1}_i)^{-1} \mathbf{1}'_i A_i \equiv M_i A_i,$$

where

$$M_i := I_{T_i} - \mathbf{1}_i (\mathbf{1}'_i \mathbf{1}_i)^{-1} \mathbf{1}'_i.$$

The matrix M_i is the within-transformation matrix for unit i . It subtracts from each observation its unit-specific time average.

Next, define the block-diagonal matrix

$$D = \text{diag}\{\mathbf{1}_1, \dots, \mathbf{1}_N\},$$

and let

$$M_D = I_n - D(D'D)^{-1}D' = \text{diag}\{M_1, \dots, M_N\}.$$

Then, for any stacked sample vector $A = (A'_1, \dots, A'_N)'$, we have

$$M_D A = \ddot{A} = \begin{pmatrix} \ddot{A}_1 \\ \vdots \\ \ddot{A}_N \end{pmatrix}.$$

A key property is that

$$M_i \mathbf{1}_i = 0.$$

Therefore, applying M_i to the fixed effects model

$$Y_i = X_i \beta + \mathbf{1}_i u_i + \varepsilon_i$$

yields

$$\ddot{Y}_i = \ddot{X}_i \beta + \ddot{\varepsilon}_i,$$

because the time-invariant component $\mathbf{1}_i u_i$ is annihilated by the within transformation.

5.12 Fixed Effects Estimator

The fixed effects estimator, also called the *within estimator*, is the OLS estimator based on the transformed model:

$$\hat{\beta}_{\text{fe}} = \left(\sum_{i=1}^N \sum_{t \in S_i} \ddot{X}_{it} \ddot{X}'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t \in S_i} \ddot{X}_{it} \ddot{Y}_{it} \right).$$

Equivalently, in unit-level matrix notation,

$$\begin{aligned} \hat{\beta}_{\text{fe}} &= \left(\sum_{i=1}^N \ddot{X}'_i \ddot{X}_i \right)^{-1} \left(\sum_{i=1}^N \ddot{X}'_i \ddot{Y}_i \right) \\ &= \left(\sum_{i=1}^N X'_i M_i X_i \right)^{-1} \left(\sum_{i=1}^N X'_i M_i Y_i \right). \end{aligned}$$

Thus, FE estimation is simply least squares applied to the data after removing unit-specific means. The estimator uses only *within-unit variation over time*; all purely cross-sectional variation that is constant within unit is swept out together with u_i .

Remark. The FE estimator is algebraically equivalent to running least squares of Y_{it} on X_{it} and unit dummy variables. This is the so-called *dummy variable regression* or *least squares dummy variable* (LSDV) formulation.⁸

⁸Strictly speaking, one uses an appropriate normalization to avoid perfect multicollinearity among the unit dummies and the intercept.

Remark. The matrix notation makes transparent what FE estimation does: M_i is a projection that removes the span of $\mathbf{1}_i$, i.e. the space of time-invariant components within unit i . Therefore, FE eliminates omitted variable bias coming from additive unit-specific heterogeneity that does not vary over time.

Covariance matrix estimation

For inference with the fixed effects estimator, there are at least three options for estimating the covariance matrix.

1. *Homoskedastic variance estimator*: valid when ε_{it} is homoskedastic and serially uncorrelated.
2. *Heteroskedasticity-robust variance estimator*: valid when ε_{it} is serially uncorrelated, while allowing for heteroskedasticity.
3. *Cluster-robust variance estimator*: valid when ε_{it} may be serially correlated within unit i .

Among these, the cluster-robust estimator is the most robust. Let

$$\hat{\varepsilon}_i = \ddot{Y}_i - \ddot{X}_i \hat{\beta}_{fe}, \quad \ddot{X}' \ddot{X} = \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i.$$

Then a cluster-robust estimator of the covariance matrix is

$$\hat{V}_{fe} = a(n, N, k) (\ddot{X}' \ddot{X})^{-1} \left(\sum_{i=1}^N \ddot{X}_i' \hat{\varepsilon}_i \hat{\varepsilon}_i' \ddot{X}_i \right) (\ddot{X}' \ddot{X})^{-1},$$

where $a(n, N, k)$ is a degrees-of-freedom adjustment. Common choices include

$$a(n, N, k) = \frac{N}{N-1} \quad \text{or} \quad a(n, N, k) = \frac{N}{N-1} \frac{n-1}{n-N-k}.$$

Here N is the number of units, n is the total number of observations, and k is the number of regressors.

Remark. The cluster-robust estimator allows for arbitrary serial dependence of the transformed errors within a unit. For this reason, it is typically the default choice in panel applications with fixed effects.

First-Differencing as an Alternative

Instead of applying the within transformation, we can eliminate the unit-specific effect u_i by first-differencing. For any variable A_{it} , define the first-difference operator by

$$\Delta A_{it} = A_{it} - A_{i,t-1}.$$

Starting from the one-way error component model

$$Y_{it} = X'_{it}\beta + u_i + \varepsilon_{it},$$

applying the difference operator yields

$$\Delta Y_{it} = \Delta X'_{it}\beta + \Delta \varepsilon_{it}, \quad \Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}.$$

The effect u_i drops out because it is constant over time within unit.

The OLS estimator based on the differenced equation is called the *first-differenced estimator*, denoted by $\hat{\beta}_{fd}$. In stacked notation, it can be written as

$$\hat{\beta}_{fd} = \left(\sum_{i=1}^N \Delta X'_i \Delta X_i \right)^{-1} \left(\sum_{i=1}^N \Delta X'_i \Delta Y_i \right).$$

Remark. Both the fixed effects estimator and the first-differenced estimator remove the time-invariant unit effect u_i , but they do so through different transformations of the data.

Remark. The choice between $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ depends on the serial dependence structure of the idiosyncratic error.⁹

1. If $T = 2$, then $\hat{\beta}_{fe} = \hat{\beta}_{fd}$.
2. The fixed effects estimator is typically more efficient when the idiosyncratic errors are i.i.d.
3. The first-differenced estimator is typically more efficient when the idiosyncratic errors behave more like a random walk.

5.13 Asymptotic Properties of the FE Estimator

To state the asymptotic results for the fixed effects estimator, it is convenient to assume, for simplicity, that the panel is balanced.

Assumption (FE'). Suppose Assumption FE holds. In addition, assume that:

1. The pairs $(\varepsilon_i, \ddot{X}_i)$, $i = 1, \dots, N$, are i.i.d. across i .

2.

$$Q := \mathbb{E}[\ddot{X}'_i \ddot{X}_i]$$

is positive definite.

3.

$$\mathbb{E}[\varepsilon_{it}^4] < \infty.$$

⁹See Wooldridge (2010) for a more detailed discussion.

4.

$$\mathbb{E} \left[\|X_{it}\|^4 \right] < \infty.$$

Remark. The i.i.d. assumption is imposed *across units* i , not across time within a given unit. Therefore, it still allows for serial dependence in the idiosyncratic errors within an individual unit.

Remark. The assumption that

$$Q = \mathbb{E} [\ddot{X}_i' \ddot{X}_i]$$

is positive definite guarantees that there is sufficient within-unit variation to identify β . In particular, it rules out regressors that are constant over time at the individual level, since such variables are eliminated by the within transformation.

Remark. These assumptions are often more plausible in micro panels than in macro panels. In microeconomic applications, cross-sectional independence across individuals may be a reasonable approximation, whereas in macro panels common shocks often induce substantial cross-unit dependence.

Consistency and Asymptotic Normality

Define

$$\Omega := \mathbb{E} [\ddot{X}_i' \ddot{\varepsilon}_i \ddot{\varepsilon}_i' \ddot{X}_i] .$$

Proposition (Consistency and asymptotic normality). *Suppose Assumption (FE') holds. Then, as $N \rightarrow \infty$ with T fixed,*

$$\hat{\beta}_{fe} \xrightarrow{p} \beta,$$

and

$$\sqrt{N}(\hat{\beta}_{fe} - \beta) \xrightarrow{d} \mathcal{N}(0, Q^{-1} \Omega Q^{-1}) .$$

Remark. This is a large- N , fixed- T asymptotic result. Therefore, it is most appropriate in panel settings with many units and a bounded time dimension.

Derivation of the estimation error. Starting from the within estimator,

$$\hat{\beta}_{fe} = \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{Y}_i \right),$$

and using the transformed model

$$\ddot{Y}_i = \ddot{X}_i \beta + \ddot{\varepsilon}_i,$$

we obtain

$$\begin{aligned}
\hat{\beta}_{fe} &= \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' (\ddot{X}_i \beta + \ddot{\varepsilon}_i) \right) \\
&= \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left[\left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right) \beta + \frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \right] \\
&= \beta + \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \right).
\end{aligned}$$

Hence,

$$\hat{\beta}_{fe} - \beta = \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \right).$$

Consistency. Under Assumption (FE'), the pairs $(\ddot{X}_i, \ddot{\varepsilon}_i)$ are i.i.d. across i , so by the weak law of large numbers,

$$\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \xrightarrow{p} \mathbb{E}[\ddot{X}_i' \ddot{X}_i] = Q.$$

Moreover,

$$\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \xrightarrow{p} \mathbb{E}[\ddot{X}_i' \ddot{\varepsilon}_i].$$

It therefore remains to show that

$$\mathbb{E}[\ddot{X}_i' \ddot{\varepsilon}_i] = 0.$$

Write the within-transformed regressor as

$$\ddot{X}_{it} = X_{it} - \bar{X}_i, \quad \bar{X}_i := \frac{1}{T} \sum_{s=1}^T X_{is}.$$

Then

$$\begin{aligned}
\mathbb{E}[\ddot{X}_i' \ddot{\varepsilon}_i] &= \sum_{t=1}^T \mathbb{E}[\ddot{X}_{it} \varepsilon_{it}] \\
&= \sum_{t=1}^T \mathbb{E}[(X_{it} - \bar{X}_i) \varepsilon_{it}] \\
&= \sum_{t=1}^T \mathbb{E}[X_{it} \varepsilon_{it}] - \sum_{t=1}^T \mathbb{E}[\bar{X}_i \varepsilon_{it}] \\
&= \sum_{t=1}^T \mathbb{E}[X_{it} \varepsilon_{it}] - \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \mathbb{E}[X_{is} \varepsilon_{it}].
\end{aligned}$$

By strict exogeneity,

$$\mathbb{E}[X_{is} \varepsilon_{it}] = 0 \quad \text{for all } s, t,$$

so

$$\mathbb{E}[\ddot{X}_i' \ddot{\varepsilon}_i] = 0.$$

Therefore,

$$\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \xrightarrow{p} 0,$$

and since Q is positive definite,

$$\left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Combining these results,

$$\hat{\beta}_{fe} \xrightarrow{p} \beta.$$

Asymptotic normality. Multiply the estimation error representation by $\sqrt{N}$:

$$\sqrt{N}(\hat{\beta}_{fe} - \beta) = \left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \right).$$

By the multivariate central limit theorem,

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \ddot{X}_i' \ddot{\varepsilon}_i \xrightarrow{d} \mathcal{N}(0, \Omega),$$

where

$$\Omega = \mathbf{Var}(\ddot{X}_i' \ddot{\varepsilon}_i) = \mathbb{E}[\ddot{X}_i' \ddot{\varepsilon}_i \ddot{\varepsilon}_i' \ddot{X}_i],$$

using $\mathbb{E}[\ddot{X}_i' \ddot{\varepsilon}_i] = 0$. Since

$$\left(\frac{1}{N} \sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \xrightarrow{p} Q^{-1},$$

Slutsky's theorem yields

$$\sqrt{N}(\hat{\beta}_{fe} - \beta) \xrightarrow{d} \mathcal{N}(0, Q^{-1} \Omega Q^{-1}).$$

Remark. The asymptotic variance has the usual sandwich form. If one additionally imposes serially uncorrelated and homoskedastic idiosyncratic errors after the within transformation, then Ω simplifies. In general, however, allowing arbitrary within-unit serial dependence leads naturally to cluster-robust inference.

5.14 Dynamic Panel Data Models: An Overview

Modeling Dynamics

So far, we have considered static panel data models. In many applications, however, outcomes are inherently dynamic, so it is natural to allow current outcomes to depend on their own past realizations.

A general dynamic panel specification is

$$Y_{it} = \alpha_1 Y_{i,t-1} + \cdots + \alpha_p Y_{i,t-p} + X'_{it}\beta + u_i + \varepsilon_{it},$$

where $\alpha_1, \dots, \alpha_p$ are autoregressive coefficients.

Remark. The key new feature relative to the static model is the presence of lagged dependent variables among the regressors. This allows persistence in outcomes over time, but it also creates new econometric complications.

The FE Estimator is Biased

The fixed effects estimator is biased in dynamic panel models when T is fixed.¹⁰ To illustrate the source of the problem, consider the simplest dynamic panel model,

$$Y_{it} = \alpha Y_{i,t-1} + u_i + \varepsilon_{it}.$$

Applying the within transformation yields

$$\ddot{Y}_{it} = \alpha \ddot{Y}_{i,t-1} + \ddot{\varepsilon}_{it}.$$

Although the within transformation eliminates the time-invariant effect u_i , it does not solve the endogeneity problem created by the lagged dependent variable. In particular,

$$\mathbb{E}[\ddot{Y}_{i,t-1} \ddot{\varepsilon}_{it}] \neq 0.$$

Remark. This is the fundamental difference between static and dynamic fixed effects models. In the static case, the within transformation can preserve exogeneity under strict exogeneity assumptions. In the dynamic case, however, the transformed lagged dependent variable is mechanically correlated with the transformed error term.

¹⁰This is the classical Nickell bias; see Nickell (1981).

A Simple Example

To make the bias explicit, suppose that $T = 3$. Then the differenced model has only one usable time period and can be written as

$$\Delta Y_{i3} = \alpha \Delta Y_{i2} + \Delta \varepsilon_{i3}.$$

In this special case, the FE and FD estimators are equivalent, and

$$\hat{\alpha}_{fd} = \hat{\alpha}_{fe} = \left(\frac{1}{N} \sum_{i=1}^N (\Delta Y_{i2})^2 \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \Delta Y_{i2} \Delta Y_{i3} \right).$$

Substituting the model for ΔY_{i3} ,

$$\begin{aligned} \hat{\alpha}_{fe} &= \left(\frac{1}{N} \sum_{i=1}^N (\Delta Y_{i2})^2 \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \Delta Y_{i2} (\alpha \Delta Y_{i2} + \Delta \varepsilon_{i3}) \right) \\ &= \alpha + \left(\frac{1}{N} \sum_{i=1}^N (\Delta Y_{i2})^2 \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \Delta Y_{i2} \Delta \varepsilon_{i3} \right). \end{aligned}$$

Hence,

$$\hat{\alpha}_{fe} - \alpha = \left(\frac{1}{N} \sum_{i=1}^N (\Delta Y_{i2})^2 \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \Delta Y_{i2} \Delta \varepsilon_{i3} \right).$$

Under standard regularity conditions,

$$\hat{\alpha}_{fe} - \alpha \xrightarrow{p} \frac{\mathbb{E}[\Delta Y_{i2} \Delta \varepsilon_{i3}]}{\mathbb{E}[(\Delta Y_{i2})^2]}.$$

The probability limit is not zero because

$$\mathbb{E}[\Delta Y_{i2} \Delta \varepsilon_{i3}] = -\mathbb{E}[Y_{i2} \varepsilon_{i2}] = -\mathbf{Var}(\varepsilon_{i2}) \neq 0.$$

To see this more explicitly, expand the covariance term:

$$\begin{aligned} \mathbb{E}[\Delta Y_{i2} \Delta \varepsilon_{i3}] &= \mathbb{E}[(Y_{i2} - Y_{i1})(\varepsilon_{i3} - \varepsilon_{i2})] \\ &= \mathbb{E}[Y_{i2} \varepsilon_{i3}] - \mathbb{E}[Y_{i2} \varepsilon_{i2}] - \mathbb{E}[Y_{i1} \varepsilon_{i3}] + \mathbb{E}[Y_{i1} \varepsilon_{i2}]. \end{aligned}$$

Under the usual orthogonality conditions, the first, third, and fourth terms are zero, so

$$\mathbb{E}[\Delta Y_{i2} \Delta \varepsilon_{i3}] = -\mathbb{E}[Y_{i2} \varepsilon_{i2}].$$

Now, from

$$Y_{i2} = \alpha Y_{i1} + u_i + \varepsilon_{i2},$$

we get

$$\mathbb{E}[Y_{i2}\varepsilon_{i2}] = \alpha\mathbb{E}[Y_{i1}\varepsilon_{i2}] + \mathbb{E}[u_i\varepsilon_{i2}] + \mathbb{E}[\varepsilon_{i2}^2] = \mathbf{Var}(\varepsilon_{i2}),$$

and therefore

$$\mathbb{E}[\Delta Y_{i2}\Delta\varepsilon_{i3}] = -\mathbf{Var}(\varepsilon_{i2}).$$

Remark. This example shows that the bias is not a small-sample accident. Even as $N \rightarrow \infty$, with T fixed, the FE estimator does not converge to the true autoregressive coefficient α .

5.15 Dynamic Panel Data Estimators

The main idea of dynamic panel data estimators is to use past values of the outcome as instrumental variables for endogenous transformed lagged dependent variables.

The Anderson–Hsiao Idea

Consider again the simple dynamic panel model

$$Y_{it} = \alpha Y_{i,t-1} + u_i + \varepsilon_{it}.$$

As shown above, the FE and FD estimators are biased when T is fixed because the transformed lagged dependent variable is correlated with the transformed error term.

A natural response is to use an instrument. In the simple case with $T = 3$, first-differencing gives

$$\Delta Y_{i3} = \alpha \Delta Y_{i2} + \Delta \varepsilon_{i3}.$$

The Anderson–Hsiao estimator uses Y_{i1} as an instrument for ΔY_{i2} . The resulting IV estimator is

$$\hat{\alpha}_{iv} = \left(\sum_{i=1}^N Y_{i1} \Delta Y_{i2} \right)^{-1} \left(\sum_{i=1}^N Y_{i1} \Delta Y_{i3} \right).$$

Substituting the differenced model,

$$\begin{aligned} \hat{\alpha}_{iv} &= \left(\sum_{i=1}^N Y_{i1} \Delta Y_{i2} \right)^{-1} \left(\sum_{i=1}^N Y_{i1} (\alpha \Delta Y_{i2} + \Delta \varepsilon_{i3}) \right) \\ &= \alpha + \left(\sum_{i=1}^N Y_{i1} \Delta Y_{i2} \right)^{-1} \left(\sum_{i=1}^N Y_{i1} \Delta \varepsilon_{i3} \right). \end{aligned}$$

Therefore, under standard regularity conditions,

$$\hat{\alpha}_{iv} \xrightarrow{p} \alpha$$

provided the instrument is valid. Here validity follows from

$$\mathbb{E}[Y_{i1} \Delta \varepsilon_{i3}] = \mathbb{E}[Y_{i1} \varepsilon_{i3}] - \mathbb{E}[Y_{i1} \varepsilon_{i2}] = 0.$$

Remark. The logic is simple: although ΔY_{it} is endogenous in the differenced equation, sufficiently old lags of Y can be orthogonal to the current differenced error and therefore serve as valid instruments.

Remark. More sophisticated dynamic panel estimators build on this same principle: they exploit lagged levels, and in some cases lagged differences, as internal instruments.

5.16 Estimating a Simple Investment Model

We now revisit a simple investment equation, following the empirical illustration in Hansen (2022). The underlying dataset is `Invest1993`, which contains information on 1962 firms observed between 1960 and 1991.

The model of interest is

$$I_{it} = \beta_1 Q_{i,t-1} + \beta_2 D_{i,t-1} + \beta_3 CF_{i,t-1} + \beta_4 T_i + e_{it},$$

where

- I : investment/assets,
- Q : market value/assets,
- D : long-term debt/assets,
- CF : cash flow/assets,
- T : indicator for stock being traded on NYSE/AMEX.

Remark. Under q -theory, one expects $\beta_1 > 0$, while the direct roles for debt and cash flow should be limited once investment opportunities are adequately controlled for by Q . By contrast, theories of financing frictions often suggest $\beta_2 < 0$ and $\beta_3 > 0$.

Data Features

The panel is highly unbalanced. The number of observations per firm ranges from 4 to 28, with mean 13. All specifications include 19 industry-code indicators.

Remark. The coefficient on T_i is not reported in the fixed effects regression because time-invariant regressors are eliminated by the within transformation.

Remark. Relative to pooled estimation, the fixed effects estimates show a smaller coefficient on Q , a more negative coefficient on D , and a larger coefficient on CF . This suggests that controlling for time-invariant firm heterogeneity matters quantitatively for the estimated investment equation.

	Pooled (robust SE)	Pooled (cluster-robust SE)	FE (cluster-robust SE)
$\hat{\beta}_1$	0.0023955 (0.0007773)	0.0023955 (0.0009802)	0.0017180 (0.0008296)
$\hat{\beta}_2$	-0.0095633 (0.0026861)	-0.0095633 (0.0040861)	-0.0138576 (0.0049152)
$\hat{\beta}_3$	0.0261072 (0.0068746)	0.0261072 (0.0111475)	0.0490859 (0.0132478)
$\hat{\beta}_4$	-0.0166869 (0.0012722)	-0.0166869 (0.0023802)	—

Table 4: Investment equation estimates

6 M-Estimators

Remark. Main references for this section are Hayashi (2000) and Hansen (2022).

6.1 Introduction

So far, the consistency and asymptotic normality results studied in the course have been derived for estimators with explicit algebraic expressions, such as OLS, IV/2SLS, and linear GMM.

The focus of this section is on nonlinear estimators for which such explicit algebraic formulas are generally not available. These estimators typically have to be computed numerically.

In particular, we study the class of *M-estimators*, which includes many important nonlinear estimation methods, such as:

1. nonlinear least squares,
2. maximum likelihood,
3. quantile regression.

6.2 Setup

Let $\rho(Y, X; \theta)$ be a criterion function. An M-estimator $\hat{\theta}$ is defined as

$$\hat{\theta} = \arg \min_{\theta \in \Theta} S_n(\theta), \quad S_n(\theta) = \frac{1}{n} \sum_{i=1}^n \rho(Y_i, X_i; \theta),$$

where

- (Y_i, X_i) denotes the observed data,
- θ is the parameter of interest,
- Θ is the parameter space,

- $S_n(\theta)$ is the sample objective function.

We will sometimes use the shorthand notation

$$\rho_i(\theta) := \rho(Y_i, X_i; \theta).$$

This framework nests OLS as a special case with

$$\rho_i(\theta) = (Y_i - X_i' \theta)^2.$$

Our main interest is in settings where no explicit algebraic expression for $\hat{\theta}$ is available, so the estimator must be characterized indirectly through the optimization problem above.

6.3 Examples of M-Estimators

Example 1: Nonlinear Least Squares

Consider the nonlinear regression model

$$Y = m(X, \theta) + e, \quad \mathbb{E}[e | X] = 0.$$

A nonlinear least squares estimator chooses θ to minimize the average squared residual,

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n (Y_i - m(X_i, \theta))^2.$$

Thus, nonlinear least squares is an M-estimator with criterion

$$\rho_i(\theta) = (Y_i - m(X_i, \theta))^2.$$

An important example is estimation of a CES production function. Let X_1 and X_2 denote two inputs. Output Y is given by

$$Y = A(\alpha X_1^\rho + (1 - \alpha) X_2^\rho)^{1/\rho}, \quad \rho \neq 0,$$

and

$$Y = A(X_1^\alpha X_2^{1-\alpha})^\nu, \quad \rho = 0.$$

Taking logs yields the nonlinear least squares specification

$$\log Y = \beta + \frac{\nu}{\rho} \log(\alpha X_1^\rho + (1 - \alpha) X_2^\rho) + e,$$

where $\beta = \log A$.

Remark. This example shows why the M-estimation framework is useful: even when the model is nonlinear in the parameters, estimation can still be formulated as minimization of

a sample criterion function.

Example 2: Maximum Likelihood and Probit

Maximum likelihood also fits into the M-estimation framework by setting

$$\rho_i(\theta) = -\log f(Y_i | X_i, \theta),$$

where $f(Y_i | X_i, \theta)$ is the conditional density or probability mass function.

Hence, the maximum likelihood estimator can be written as

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \rho_i(\theta) = \arg \min_{\theta \in \Theta} \left(-\frac{1}{n} \sum_{i=1}^n \log f(Y_i | X_i, \theta) \right),$$

which is equivalent to maximizing the sample log-likelihood.

A prominent example is the Probit model for a binary outcome:

$$P(Y = 1 | X) = \Phi(X'\theta),$$

where $\Phi(\cdot)$ is the standard normal distribution function.

For binary $Y_i \in \{0, 1\}$, the corresponding M-estimation criterion is

$$\rho_i(\theta) = -Y_i \log(\Phi(X_i'\theta)) - (1 - Y_i) \log(1 - \Phi(X_i'\theta)).$$

Remark. This is simply the negative log-likelihood contribution of observation i . Thus, maximum likelihood is not a separate estimation logic from M-estimation; it is one important special case of it.

Example 3: Quantile Regression

Suppose that instead of modeling the conditional mean $\mathbb{E}[Y | X]$, we are interested in modeling the conditional τ -quantile

$$Q_{Y|X}(\tau | X), \quad \tau \in (0, 1).$$

A common specification imposes the linear model

$$Q_{Y|X}(\tau | X) = X'\theta(\tau).$$

Linear quantile regression fits into the M-estimation framework with criterion

$$\rho_i(\theta) = (Y_i - X_i'\theta) \left(\tau - \mathbf{1}\{Y_i - X_i'\theta < 0\} \right).$$

Equivalently, if we define the residual $u_i(\theta) = Y_i - X_i'\theta$, then

$$\rho_i(\theta) = \rho_\tau(u_i(\theta)), \quad \rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}),$$

where $\rho_\tau(\cdot)$ is the *check function*.

Remark. Quantile regression was introduced by Koenker and Bassett (1978). Unlike least squares, it targets a conditional quantile rather than the conditional mean, making it useful when the effects of regressors differ across the outcome distribution.

6.4 Consistency

Identification

To study consistency in the M-estimation framework, we first introduce the corresponding population objective function. Define

$$S(\theta) := \mathbb{E}[S_n(\theta)] = \mathbb{E}[\rho(Y, X; \theta)],$$

assuming

$$\mathbb{E}[|\rho(Y, X; \theta)|] < \infty \quad \text{for each } \theta \in \Theta.$$

We then define the population parameter

$$\theta_0 := \arg \min_{\theta \in \Theta} S(\theta).$$

Following Hansen (2022), we say that θ is *point identified* if θ_0 is unique.

Remark. The sample estimator $\hat{\theta}$ minimizes the sample objective $S_n(\theta)$, while the target parameter θ_0 minimizes the population objective $S(\theta)$. Thus, consistency requires that the sample criterion approximate the population criterion well enough for their minimizers to be close.

Remark. In these notes we take identification as given. In applications, however, uniqueness of θ_0 must be verified on a case-by-case basis.

Why Pointwise Convergence Is Not Enough

By the weak law of large numbers, for any fixed θ ,

$$S_n(\theta) \xrightarrow{p} S(\theta).$$

However, pointwise convergence alone is not sufficient to conclude that

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

The reason is that $\hat{\theta}$ is itself random and depends on the entire shape of the criterion function over Θ , not just on its value at one fixed parameter. Thus, to control the behavior of the minimizer, we need a stronger notion of convergence.

Remark. Pointwise convergence says that for each fixed θ , the number $S_n(\theta)$ gets close to $S(\theta)$. But minimization is a global operation over the whole parameter space. Therefore, consistency of the minimizer typically requires the sample objective to be uniformly close to the population objective.

Uniform Convergence in Probability

We say that $S_n(\theta)$ converges in probability to $S(\theta)$ *uniformly* over $\theta \in \Theta$ if

$$\sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| \xrightarrow{p} 0.$$

This is much stronger than pointwise convergence. It guarantees that the entire sample objective function lies uniformly close to the population objective function over the whole parameter space.

Remark. Uniform convergence is the key bridge from convergence of objective functions to convergence of argmin estimators.

Uniform Law of Large Numbers

Theorem 1 (Uniform Law of Large Numbers). *Suppose that:*

1. $(Y_i, X_i), i = 1, \dots, n$, are i.i.d.;
2. $\rho(Y, X; \theta)$ is continuous in $\theta \in \Theta$ with probability one;
3. there exists a dominating function $G(Y, X)$ such that

$$|\rho(Y, X; \theta)| \leq G(Y, X) \quad \text{for all } \theta \in \Theta,$$

and

$$\mathbb{E}[G(Y, X)] < \infty;$$

4. Θ is compact.

Then

$$\sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| \xrightarrow{p} 0.$$

Remark. This result is the M-estimation analogue of the usual law of large numbers, but strengthened to hold uniformly over the parameter space.

Remark. The role of the assumptions is intuitive. Continuity in θ prevents the criterion from oscillating too wildly across nearby parameter values; the envelope condition provides integrable control of the criterion; and compactness of Θ rules out pathological behavior “at infinity.”

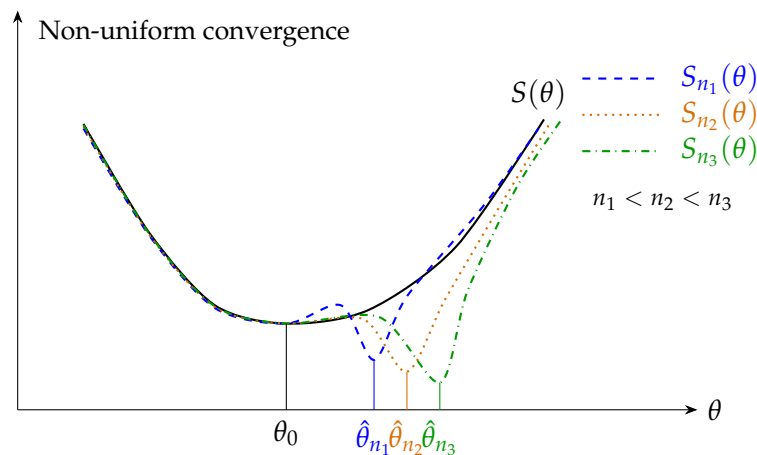


Figure 1: Pointwise convergence of $S_n(\theta)$ to $S(\theta)$ does not by itself imply convergence of the minimizers.

Remark. In this figure, $S_n(\theta) \xrightarrow{p} S(\theta)$ for every fixed θ , but the minimizers $\hat{\theta}_n$ do not converge to θ_0 . This illustrates why pointwise convergence is not enough and motivates the need for uniform convergence.

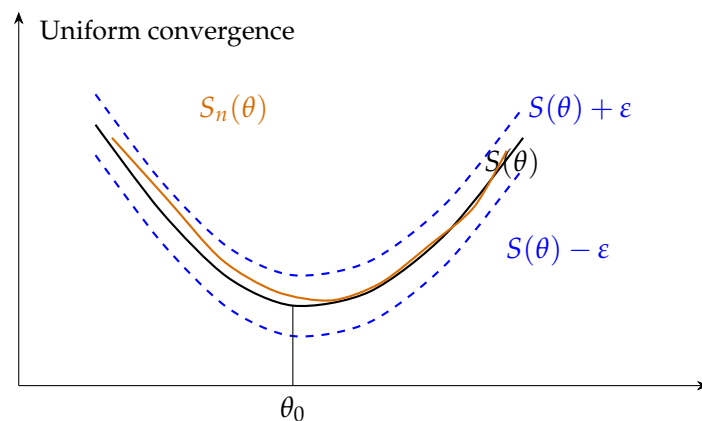


Figure 2: Illustration of uniform convergence: the sample criterion $S_n(\theta)$ lies uniformly within an ε -band around $S(\theta)$.

Remark. This figure illustrates uniform convergence in probability: for n sufficiently large, the sample objective $S_n(\theta)$ is uniformly close to the population objective $S(\theta)$ over all $\theta \in \Theta$, in the sense that

$$\sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| < \varepsilon.$$

Consistency of M-Estimators

Proposition (Consistency under high-level conditions). *Suppose that:*

1. $S_n(\theta)$ converges in probability to $S(\theta)$ uniformly over $\theta \in \Theta$, that is,

$$\sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| \xrightarrow{p} 0;$$

2. for every $\varepsilon > 0$,

$$\inf_{\|\theta - \theta_0\| \geq \varepsilon} S(\theta) > S(\theta_0).$$

Then

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

Proof. We proceed in two steps.

Step 1: Show that $S(\hat{\theta}) - S(\theta_0) \xrightarrow{p} 0$. Since θ_0 minimizes $S(\theta)$, we have

$$S(\theta_0) \leq S(\hat{\theta}).$$

Since $\hat{\theta}$ minimizes $S_n(\theta)$, we also have

$$S_n(\hat{\theta}) \leq S_n(\theta_0).$$

Therefore,

$$\begin{aligned} 0 &\leq S(\hat{\theta}) - S(\theta_0) \\ &= (S(\hat{\theta}) - S_n(\hat{\theta})) + (S_n(\hat{\theta}) - S(\theta_0)) \\ &\leq (S(\hat{\theta}) - S_n(\hat{\theta})) + (S_n(\theta_0) - S(\theta_0)) \\ &\leq \sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| + \sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)|. \end{aligned}$$

Hence,

$$0 \leq S(\hat{\theta}) - S(\theta_0) \leq 2 \sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| \xrightarrow{p} 0,$$

which proves

$$S(\hat{\theta}) - S(\theta_0) \xrightarrow{p} 0.$$

Step 2: Use the separation condition. Fix any $\varepsilon > 0$. By assumption (ii),

$$\inf_{\|\theta - \theta_0\| \geq \varepsilon} S(\theta) > S(\theta_0).$$

Define

$$\delta_\varepsilon := \inf_{\|\theta - \theta_0\| \geq \varepsilon} S(\theta) - S(\theta_0) > 0.$$

If $\|\hat{\theta} - \theta_0\| \geq \varepsilon$, then necessarily

$$S(\hat{\theta}) - S(\theta_0) \geq \delta_\varepsilon.$$

Thus,

$$P(\|\hat{\theta} - \theta_0\| \geq \varepsilon) \leq P(S(\hat{\theta}) - S(\theta_0) \geq \delta_\varepsilon).$$

By Step 1, the probability on the right converges to zero. Since $\varepsilon > 0$ was arbitrary,

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

□

Remark. This is a high-level argmin consistency result. The key ingredients are:

1. the sample criterion must approximate the population criterion uniformly;
2. the population criterion must separate θ_0 from all other parameter values.

Corollary (Consistency under ULLN conditions). Suppose that the conditions of the Uniform Law of Large Numbers hold and that θ_0 uniquely minimizes $S(\theta)$. Then

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

Remark. The corollary simply combines the ULLN with the argmin consistency proposition. In this way, low-level primitive assumptions imply consistency of the M-estimator.

6.5 Asymptotic Normality

We now turn to a general asymptotic normality result for M-estimators.

Let $\mathcal{N}$ denote a neighborhood of θ_0 . Define

$$\psi(Y, X; \theta) := \frac{\partial}{\partial \theta} \rho(Y, X; \theta), \quad \psi_i(\theta) := \psi(Y_i, X_i; \theta),$$

and

$$\bar{\psi}_n(\theta) := \frac{\partial}{\partial \theta} S_n(\theta) = \frac{1}{n} \sum_{i=1}^n \psi_i(\theta).$$

Also define

$$\psi_i := \psi_i(\theta_0), \quad \psi(\theta) := \frac{\partial}{\partial \theta} S(\theta) = \mathbb{E}[\psi(Y, X; \theta)],$$

and

$$Q(\theta) := \frac{\partial^2}{\partial \theta \partial \theta'} S(\theta), \quad Q := Q(\theta_0).$$

Theorem 2 (Asymptotic normality of M-estimators). *Suppose that the assumptions of the previous consistency proposition hold. In addition, assume that:*

1.

$$\mathbb{E}[\|\psi(Y, X; \theta_0)\|^2] < \infty;$$

2. Q is positive definite;

3. $Q(\theta)$ is continuous in $\theta \in \mathcal{N}$;

4. for all $\theta_1, \theta_2 \in \mathcal{N}$,

$$\|\psi(Y, X; \theta_1) - \psi(Y, X; \theta_2)\| \leq B(Y, X)\|\theta_1 - \theta_2\|,$$

where

$$\mathbb{E}[B(Y, X)^2] < \infty;$$

5. $\theta_0 \in \text{int}(\Theta)$.

Then, as $n \rightarrow \infty$,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, Q^{-1}\Omega Q^{-1}), \quad \Omega := \mathbb{E}[\psi_i \psi_i'].$$

Remark. The asymptotic covariance matrix has the usual sandwich form. The matrix Q captures the local curvature of the population criterion, while Ω captures the variance of the score-like term ψ_i .

Proof Sketch

Since $\theta_0 \in \text{int}(\Theta)$ and $\hat{\theta} \xrightarrow{p} \theta_0$, the estimator is eventually interior with probability approaching one, so it satisfies the first-order condition

$$\bar{\psi}_n(\hat{\theta}) = 0.$$

Apply a first-order Taylor expansion of $\bar{\psi}_n(\hat{\theta})$ around θ_0 :

$$0 = \bar{\psi}_n(\theta_0) + \left(\frac{\partial}{\partial \theta'} \bar{\psi}_n(\tilde{\theta}) \right) (\hat{\theta} - \theta_0),$$

where $\tilde{\theta}$ lies between $\hat{\theta}$ and θ_0 . Rearranging,

$$\sqrt{n}(\hat{\theta} - \theta_0) = - \left(\frac{\partial}{\partial \theta'} \bar{\psi}_n(\tilde{\theta}) \right)^{-1} \sqrt{n} \bar{\psi}_n(\theta_0).$$

Now study the two terms separately.

Term A: Hessian part. Because

$$\frac{\partial}{\partial \theta'} \bar{\psi}_n(\tilde{\theta}) = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta \partial \theta'} \rho(Y_i, X_i; \tilde{\theta}),$$

the law of large numbers together with continuity implies

$$\frac{\partial}{\partial \theta'} \bar{\psi}_n(\tilde{\theta}) \xrightarrow{p} Q.$$

Hence,

$$\left(\frac{\partial}{\partial \theta'} \bar{\psi}_n(\tilde{\theta}) \right)^{-1} \xrightarrow{p} Q^{-1}.$$

Term B: Score part. At θ_0 ,

$$\sqrt{n} \bar{\psi}_n(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i.$$

Because θ_0 minimizes $S(\theta)$, it satisfies the population first-order condition

$$\psi(\theta_0) = \mathbb{E}[\psi_i] = 0.$$

Therefore, by the multivariate central limit theorem,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i \xrightarrow{d} \mathcal{N}(0, \Omega), \quad \Omega = \mathbb{E}[\psi_i \psi_i'].$$

Combining the two parts and applying Slutsky's theorem,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} -Q^{-1} \mathcal{N}(0, \Omega) = \mathcal{N}(0, Q^{-1} \Omega Q^{-1}).$$

Remark. The minus sign disappears in the asymptotic variance. The result is therefore exactly the familiar sandwich formula:

$$\mathbf{Var}(\sqrt{n}(\hat{\theta} - \theta_0)) = Q^{-1} \Omega Q^{-1}.$$

7 Appendix: Econometric Pearls

7.1 Pearl 1: OLS as a Difference in Means

A useful way to think about OLS is that it chooses coefficients so that the residuals cannot be systematically predicted by the regressors. In the simplest treatment-control case, this intuition becomes especially transparent.

Consider the regression

$$Y_i = \alpha + \beta D_i + e_i, \quad D_i \in \{0, 1\},$$

where $D_i = 1$ denotes treated units and $D_i = 0$ denotes control units. The OLS moment conditions are

$$\sum_{i=1}^n \hat{e}_i = 0, \quad \sum_{i=1}^n D_i \hat{e}_i = 0.$$

The second condition says that residuals must sum to zero among treated units:

$$\sum_{i:D_i=1} (Y_i - \hat{\alpha} - \hat{\beta}) = 0.$$

Hence,

$$\hat{\alpha} + \hat{\beta} = \frac{1}{n_1} \sum_{i:D_i=1} Y_i = \bar{Y}_1.$$

The first condition says that total residuals must sum to zero. Since residuals already sum to zero among treated units, they must also sum to zero among control units:

$$\sum_{i:D_i=0} (Y_i - \hat{\alpha}) = 0.$$

Therefore,

$$\hat{\alpha} = \frac{1}{n_0} \sum_{i:D_i=0} Y_i = \bar{Y}_0.$$

Combining the two equations gives

$$\hat{\beta}_{\text{OLS}} = \bar{Y}_1 - \bar{Y}_0.$$

Thus, with a constant and a binary regressor, OLS is exactly a difference in sample means. The fitted value for the control group is the control-group mean, and the fitted value for the treatment group is the treatment-group mean:

$$\hat{Y}(0) = \bar{Y}_0, \quad \hat{Y}(1) = \bar{Y}_1.$$

The coefficient on D_i is simply the vertical distance between these two group means.

The same logic holds in population. The moment conditions are

$$\mathbb{E}[e_i] = 0, \quad \mathbb{E}[D_i e_i] = 0.$$

Since D_i is binary,

$$\mathbb{E}[D_i e_i] = \mathbb{P}(D_i = 1) \mathbb{E}[e_i \mid D_i = 1].$$

If $\mathbb{P}(D_i = 1) > 0$, then

$$\mathbb{E}[e_i \mid D_i = 1] = 0.$$

Together with $\mathbb{E}[e_i] = 0$, this implies

$$\mathbb{E}[e_i \mid D_i = 0] = 0.$$

Therefore,

$$\alpha + \beta = \mathbb{E}[Y_i \mid D_i = 1], \quad \alpha = \mathbb{E}[Y_i \mid D_i = 0],$$

and hence

$$\beta = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0].$$

The important distinction is between the algebraic interpretation and the causal interpretation. Algebraically, OLS with a binary regressor estimates a difference in observed means. Causally, this difference is an average treatment effect only under additional assumptions, such as random assignment:

$$(Y_i(1), Y_i(0)) \perp D_i.$$

Without such an assumption, the observed difference decomposes as

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{effect on the treated}} + \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}$$

The econometric pearl is therefore:

OLS is a difference in means when the regressor is binary.

More generally, OLS makes residuals orthogonal to the regressors. With a binary regressor, orthogonality means that residuals average to zero within treatment and control groups. With continuous regressors, the same idea becomes a slope: OLS chooses the best linear contrast so that no linear information about the residual remains in the regressors.

7.2 Pearl 2: GMM in the Overidentified Case

Consider moment conditions of the form

$$\mathbb{E}[g_i(\beta_0)] = 0,$$

where $g_i(\beta) \in \mathbb{R}^\ell$ and $\beta \in B \subseteq \mathbb{R}^k$. The model is *overidentified* when

$$\ell > k.$$

In this case, there are more moment restrictions than parameters. Therefore, in finite samples, it is generally impossible to find a value of β such that

$$\bar{g}_n(\beta) = 0, \quad \bar{g}_n(\beta) = \frac{1}{n} \sum_{i=1}^n g_i(\beta).$$

GMM replaces the problem of solving all moment equations exactly with the problem of making the sample moments as close to zero as possible.

Let W be an $\ell \times \ell$ symmetric positive definite matrix. The GMM criterion is

$$Q_n(\beta) = \bar{g}_n(\beta)' W \bar{g}_n(\beta),$$

or, equivalently,

$$J_n(\beta) = n \bar{g}_n(\beta)' W \bar{g}_n(\beta).$$

The factor n does not affect the minimizer. It is included because it gives the J -statistic a non-degenerate asymptotic distribution.

The GMM estimator is

$$\hat{\beta}_{GMM} = \arg \min_{\beta \in B} Q_n(\beta).$$

The population counterpart of the sample moment is

$$g(\beta) = \mathbb{E}[g_i(\beta)].$$

Hence, the population GMM criterion is

$$Q(\beta) = g(\beta)' W g(\beta).$$

Since W is positive definite,

$$Q(\beta) \geq 0$$

for all β , and

$$Q(\beta) = 0 \iff g(\beta) = 0.$$

If the moment restrictions identify β_0 , that is,

$$g(\beta) = 0 \iff \beta = \beta_0,$$

then

$$Q(\beta)$$

has a unique minimum at β_0 .

Now suppose that a uniform law of large numbers holds:

$$\sup_{\beta \in B} \|\bar{g}_n(\beta) - g(\beta)\| \xrightarrow{p} 0.$$

Then the sample criterion converges uniformly to the population criterion:

$$\sup_{\beta \in B} |Q_n(\beta) - Q(\beta)| \xrightarrow{p} 0.$$

Therefore, by the argmin consistency theorem,

$$\hat{\beta}_{GMM} \xrightarrow{p} \beta_0.$$

The key idea is that overidentified GMM cannot generally make all sample moments equal to zero. Instead, it chooses the value of β that makes the vector of sample moments closest to zero under the quadratic metric defined by W . The role of W is not identification. Identification comes from the population condition

$$g(\beta) = 0 \iff \beta = \beta_0.$$

The role of W is to determine how different moment conditions are traded off when they cannot all be satisfied exactly in sample.

Later, efficiency considerations determine the optimal choice of W . If

$$\Omega = \mathbb{E}[g_i(\beta_0)g_i(\beta_0)'],$$

then the efficient weighting matrix is

$$W^* = \Omega^{-1}.$$

Thus, moments with higher variance receive less weight, and correlated moments are not counted as independent sources of information.